

What Adapter Ensembling Actually Buys: A Pre-Registered Audit Against a Stronger Single Adapter

Anonymous ACL submission

Abstract

Parameter-efficient fine-tuning leaves practitioners with populations of adapters, and a large literature reports that combining them helps. We ask what combining them buys against a strong single adapter, in two steps. First, we audit a 1,028-cell corpus (67 pools, 10 base models, 8 tasks) registered at git SHA 02a5a6bc *before any adapter was trained*¹, adjudicating each (pool, rule, baseline, metric) cell with paired CIs and per-arm FDR control. Against a better-resourced single, encoder ensembling never significantly improves accuracy (0 of 92 cells; 29% significantly worse) and decoder ensembling sometimes does (13 of 100, all MNLI), though pools sharing baselines leave only 10 and 15 independent clusters. Second, a pre-registered experiment on two tasks the corpus never used, with exactly matched training budgets and sealed test sets, confirms the split: at every budget an ensemble of rank-8 adapters loses to the best rank-32 single on BERT-base (−1.0 to −2.8pp) and beats it on Qwen2.5-0.5B (+1.0 to +3.2pp), and the decoder ensemble keeps a lower NLL after both sides are temperature-scaled. A single that tunes rank and learning rate inside the same budget narrows the decoder gap but does not close it. An 8-member ensemble costs 11–18× a merged single’s measured latency. Comparing against the pool’s best member instead of the stronger single triples the SUPPORTED rate (6.8% → 21.9% on the same 48 pools). We release the cells, the experiment’s member logits and adapters, and code to evaluate new combination rules on them.

¹Cell count: 1,028 = 827 v1.0 main-pool cells (55 pools × ensemble methods × {best_of_n, n_rank} × {accuracy, ECE}) + 100 v1.1 HellaSwag multichoice cells + 100 baseline self-comparison cells + 1 Phase-0 smoke-test cell. The 2 v1.1 GSM8K and 6 v1.2 supplementary pools (App. K) are reported separately in Table 9 and are not in the headline 1,028.

1 Introduction

Parameter-efficient fine-tuning (PEFT) with LoRA-style adapters (Hu et al., 2022) has made it routine to keep dozens of task-specific adapters around: serving stacks such as vLLM and S-LoRA (Kwon et al., 2023; Sheng et al., 2024) route per request to one of N adapters at negligible extra cost, and frameworks such as LoRAHub (Huang et al., 2024) target multi-adapter composition directly. The practitioner’s question is whether *combining* those adapters — averaging weights, voting on labels, or routing through them — beats spending the same budget on a single adapter. A large literature reports that it does. We ask a prior question: *how much of that reported benefit survives a strict evaluation protocol?*

Answering this requires comparisons fixed before the data was seen, because the choices involved — which single adapter counts as the baseline, how it was selected, at what budget, on which benchmark — are the choices that determine the answer. We take two steps. First, an *audit*: we pre-registered ADAPTERFRONTIER at git SHA 02a5a6bc before any adapter was trained and adjudicate each (pool, rule, baseline, metric) cell with paired bootstrap CIs and per-arm Benjamini–Hochberg FDR (Benjamini and Hochberg, 1995). Second, because the audit’s baseline is better-resourced rather than budget-matched and its member logits were not kept, a *pre-registered experiment* that matches training budgets exactly, calibrates both sides and is scored on sealed test sets.

What the audit finds. Its `n_rank` single received a median $1.45\times$ the pool’s training compute plus a hyperparameter search, so it is a better-resourced competitor, not a matched one. Against it, **no encoder cell shows a significant accuracy gain** (0 of 92) while 29% are significantly worse; on decoders 13 of 100 cells are SUPPORTED, none REVERSED, all on MNLI. These counts over-

state the independent evidence: pools sharing a baseline and test set leave 10 encoder and 15 decoder clusters, at which level the encoder mean is -1.20pp (95% CI $[-2.30, -0.37]$) and the decoder mean $+0.66\text{pp}$ ($[-0.09, +1.37]$). Ensembles improve ECE in 51 of the same 92 encoder cells, but against a single that was never calibrated, and the corpus kept too few member logits to supply the control (§4).

What the experiment finds. On SNLI and Yahoo Answers, neither in the corpus, at budgets of 4–16 epochs with identical training tokens on both sides, an ensemble of rank-8 adapters loses to the best rank-32 single on BERT-base (-1.0 to -2.8pp) and beats it on Qwen2.5-0.5B ($+1.0$ to $+3.2\text{pp}$), as registered; after one temperature per arm the decoder ensemble’s NLL is still lower and the encoder ensemble’s higher. Letting the single tune rank and learning rate inside the same budget leaves BERT unchanged and narrows the Qwen gap to $+1.1/+1.7\text{pp}$; no Qwen configuration reaches the ensemble even when picked on test (§5). The decoder gain costs $11\text{--}18\times$ a merged single’s measured latency at $N=8$.

Three shortcuts move the verdict. Selecting the “best single” on test is worth $+1.12\text{pp}$ to whichever arm may peek. On the same 48 pools, comparing against the pool’s best member rather than the stronger single triples the SUPPORTED rate ($6.8\% \rightarrow 21.9\%$) and cuts REVERSED from 14.1% to 3.6% . And MNLI’s -0.012pp task mean reads as “no effect” while concealing 30 significant cells pointing both ways (§6).

Contributions. (i) **A pre-registered corpus and a reusable subset:** 1,028 cell JSONs (927 valid) with per-cell CIs, p -values and verdicts that re-adjudicate from released predictions, and 132 new runs (the 120 of §5 plus a 12-run pilot) whose per-member validation and test logits are in the anonymous release, on which a new combination rule is evaluated end to end without retraining (examples/new_combiner_demo.py). (ii) **An audit against a better-resourced single,** with the dependence structure stated: an encoder null with frequent harm, a narrow decoder gain. (iii) **A pre-registered equal-budget test** that confirms the family split in accuracy and in calibrated NLL, survives a tuned single, and prices the gain in measured latency. (iv) **A protocol-sensitivity analysis** pricing three shortcuts that each move the

verdict (Table 1, Fig. 1).

2 Related work

PEFT and adapter ensembling. The adapter line of work begins with bottleneck adapters (Houlsby et al., 2019) and the AdapterHub framework (Pfeiffer et al., 2020); LoRA (Hu et al., 2022), DoRA (Liu et al., 2024), PiSSA (Meng et al., 2024), AdaLoRA (Zhang et al., 2023), and VeRA (Kopiczko et al., 2024) make many low-rank adapters per task practical. Combination strategies split into three families: *prediction-space* (soft/hard voting, logit averaging (Lakshminarayanan et al., 2017), including ensembles of LoRA adapters on LLMs (Wang et al., 2023a)); *weight-space* (model soups (Wortsman et al., 2022), TIES (Yadav et al., 2023), DARE (Yu et al., 2024)); and *routing / mixture-of-adapters* (AdapterFusion (Pfeiffer et al., 2021) composes via attention; LoRAHub (Huang et al., 2024) optimizes a continuous weight vector; AdaMix (Wang et al., 2022) mixes multiple LoRA experts inside training; MoLE (Wu et al., 2024) and multi-head adapter routing (Caccia et al., 2023) learn routing over adapter experts; HydraLoRA (Tian et al., 2024) uses asymmetric multi-head LoRA; MultiLoRA (Wang et al., 2023b) reduces interference among co-trained LoRAs). Our adjudication framework treats each of these methods as a *plug-in* that produces per-example test predictions and can be added to our cell corpus without retraining the pools. We implement and report prediction-space methods directly (soft_vote, logit_avg, majority_vote, greedy_soup — which greedily adds members on val_selection and averages their logits, the prediction-space analogue of a greedy soup — plus a prediction-space uniform_soup proxy that averages per-adapter logits); weight-space TIES/DARE/uniform-merge implementations live in weight_space_merge.py and are reported as a separate family in App. C. Routing or weighting methods need per-member logits, which survive only for the runs of §5 and six corpus pools (L15).

Pre-registration in ML. Pre-registration is standard in clinical trials and increasingly common in cognitive science but rare in ML (Munafò et al., 2017; Loken and Gelman, 2017); Hofman et al. (2021) argue for tighter coupling of explanation and prediction in computational social science, an argument that translates directly to empirical NLP

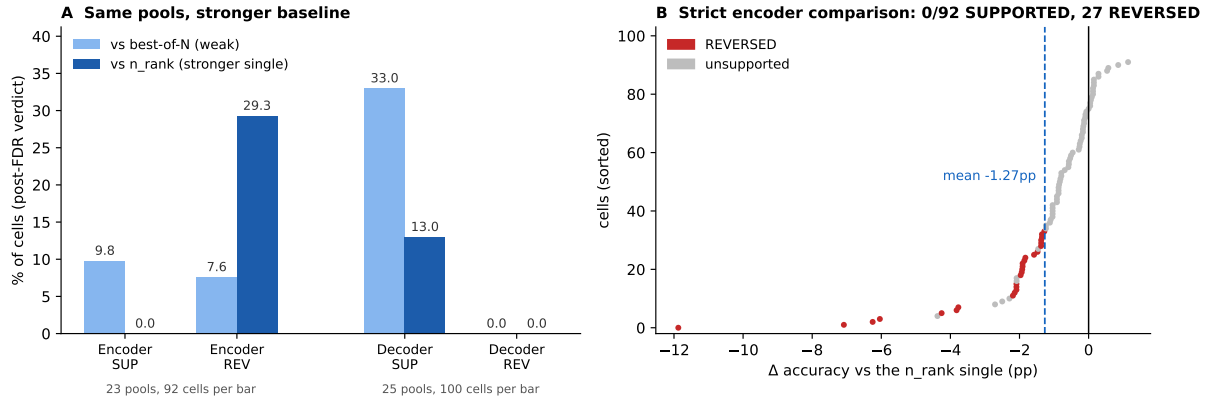


Figure 1: **The apparent ensembling benefit depends on the baseline.** (A) The same 48 pools, adjudicated against the pool’s own best member (`best_of_n`, light) and against the better-resourced `n_rank` single (dark); every pair of bars shares its cells. On encoders SUPPORTED falls from 9.8% to 0% and REVERSED rises from 7.6% to 29.3%; on decoders SUPPORTED falls from 33% to 13% and nothing is REVERSED under either baseline. (B) All 92 strict encoder cells, sorted, colored by post-FDR verdict: the mass lies left of zero (mean -1.27pp), 27 are REVERSED and none SUPPORTED.

— Bowman and Dahl (2021) make a closely related case that NLU benchmark results need pre-committed protocols if they are to drive scientific (rather than leaderboard) progress. Dehghani et al. (2022) further argue that compute-matched comparisons are essential to honest efficiency claims; the corpus’s baseline is better-resourced rather than matched, so we test matched budgets separately (§5). We adopt pre-registration here: the protocol, splits, baselines, and statistical tests were locked at git SHA 02a5a6bc before any pool was trained.

Calibration of fine-tuned LMs. ECE in its modern form was introduced by Naeini et al. (2015) and popularised with equal-width bins by Guo et al. (2017); we use 15 equal-mass bins, with refinements such as soft calibration objectives (Karandikar et al., 2021). Nixon et al. (2019) survey ECE estimator bias under varying bin counts and small- N regimes; Minderer et al. (2021) re-examine calibration of modern (post-2020) architectures and find that distribution-shift calibration does not always follow in-distribution calibration. Deep ensembles (Lakshminarayanan et al., 2017) reliably lower ECE on classifiers, but Ashukha et al. (2020) argue that in-domain uncertainty comparisons, ensembles included, should be made after temperature scaling, since an uncalibrated baseline confounds them, and Ovadia et al. (2019) find that temperature scaling’s in-distribution gains do not survive dataset shift, where ensembles hold up best. Our recalibration control applies the first point to PEFT pools; the second bounds it (L13). We extend

this measurement to PEFT pools and report Brier, NLL, MCE alongside ECE, with paired bootstrap CIs; per-token logprob ECE for generative tasks (App I) is a non-standard adaptation we explicitly document.

Efficient multi-adapter serving. S-LoRA (Sheng et al., 2024) and Punica (Chen et al., 2024) (the latter integrated in vLLM (Kwon et al., 2023)) enable thousands of LoRA adapters to share a single base-model forward; we measure the runtime cost of cell-by-cell ensembling under this serving regime.

When reported gains do not survive fair comparison. Several fields have found that a literature of reported improvements shrinks or disappears once baselines are tuned and budgets are matched: GAN variants become indistinguishable under equal hyperparameter search (Lucic et al., 2018), deep metric-learning gains are “marginal at best” under controlled evaluation (Musgrave et al., 2020), and most neural recommenders are beaten by simple heuristics (Ferrari Dacrema et al., 2019). The recurring cause is not fraud but protocol: an under-tuned or under-resourced baseline, a selection step that sees the test set, or an aggregate score that hides variance (Dodge et al., 2019; Reimers and Gurevych, 2017). NLP has documented the same failure modes for benchmarking practice (Bowman and Dahl, 2021) and for efficiency claims specifically, where comparisons that do not fix the compute budget can invert conclusions (Dehghani et al., 2022). Two literatures bear directly on our ques-

tion. PEFT evaluation has already had its reality check: [Chen et al. \(2022\)](#) find validation and testing practice across PEFT studies unreliable once method instability is accounted for. And *ensemble versus one bigger model at matched cost* is an open dispute rather than a naive assumption: [Abe et al. \(2022\)](#) find a single larger network replicates ensemble gains with diversity contributing little, [Kondratyuk et al. \(2020\)](#) conclude the opposite at matched FLOPs, [Lobacheva et al. \(2020\)](#) show the optimum depends on budget, and [Theisen et al. \(2023\)](#) predict the gain from the disagreement-to-error ratio. Our encoder result is the PEFT instance of the [Abe et al.](#) side and our decoder result leans the other way; our descriptive frontier is in line with [Theisen et al.](#). We claim priority for neither. Ours is the pre-registration before training, the cell-level adjudication, and the per-shortcut price in percentage points (§6).

Benchmarks for NLP reasoning. HellaSwag ([Zellers et al., 2019](#)), GSM8K ([Cobbe et al., 2021](#)), and the GLUE-family tasks ([Wang et al., 2018](#)) are scored by our own evaluators, not by lm-evaluation-harness ([Gao et al., 2023](#)), though we follow its length-normalised log-likelihood convention on HellaSwag. We use them as cells, not as headline numbers.

3 Benchmark composition

3.1 Pool types

Per (model, task) cell:

- **Pool-A (seed-only):** 20 LoRA adapters with fixed ($r = 8, \alpha = 16, lr = 3e-4, target = qv$), seed varied across $\{11, 21, \dots, 201\}$.
- **Pool-B (HP-diverse):** 24 adapters from $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{1e-4, 3e-4\}$.
- **Pool-C (method-mixed):** 20 adapters, $\{\text{LoRA}, \text{DoRA}, \text{rsLoRA}, \text{PiSSA}\} \times 5$ seeds.

Counting pools. The release holds 99 manifests: 73 trained ensemble pools, 25 n_rank baseline pools and one smoke test. The pre-registered corpus is 67 of the 73 (55 v1.0 classification + 12 v1.1 HellaSwag/GSM8K); the other 6 are v1.2 supplements (Table 9). Where a table counts 80 pools, it is the 55 v1.0 pools plus the 25 baselines, which also enter best_of_n comparisons. Realized sizes fall short of the design (20/24/20): median

19, range 7–24; baselines run 6–20, so best-of- N selection pressure varies by cell.

The n_rank baseline, as run. Per-pool grids are in App. J. The pre-registration specified one adapter of rank $N \times$ the pool’s; what was run is a search over $r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds, best on val_selection (Table 7). Measured from the manifests it used a median $1.45 \times$ the pool’s training GPU-hours (max $2.96 \times$). It is therefore a *better-resourced* competitor that serves one adapter against the ensemble’s N — neither budget-matched nor an upper bound on what a single adapter can reach. The equal-budget question is answered separately, by a pre-registered experiment (§5). best_of_n is the val_selection-best member of the pool itself.

3.2 Tasks and models

The in-domain set is six classification tasks: MNLI (3-way NLI, $n_{\text{test}}=3,926$), QNLI (2,186), AG News (4-way topic, 3,040), BoolQ (1,308), ANLI R3 (adversarial NLI, 480), and SST-2 (350), all evaluated on the frozen 40% test slice of the split described above. ANLI, SST-2 and BoolQ fall below $n_{\text{test}}=2,000$ and are the source of the low-precision cells quantified in §4; dropping small tasks post hoc is itself a shortcut, so we keep them and stratify by test-set size wherever the null is at stake. ANLI also serves as an OOD target for MNLI-trained pools (App. F), on disjoint pools, never pooled into one cell.

Architectures span two families. *Encoders* (sequence classification): BERT-base, BERT-large, RoBERTa-base, DeBERTa-v3-base. *Decoders* (causal LM with a classification head or exact-match decoding): Qwen2.5- $\{0.5\text{B}, 1.5\text{B}, 3\text{B}, 7\text{B}\}$, TinyLlama-1.1B, Pythia-1.4B, SmolLM2-1.7B, Llama-3.1-8B-Instruct, and Mistral-7B. Encoder and decoder cells behave differently enough (App. W) that we never pool them; §4 reports the two families side by side.

v1.1 task-family extensions. The six in-domain tasks cover classification only. v1.1 adds two output spaces — multi-choice (HellaSwag, four endings) and free-form generation (GSM8K, exact match) — through twelve further pools, bringing the pre-registered corpus to **67** pools. The extension itself added cells under the pre-registration’s amendment clause without changing any v1.0 cell (later corrections to the statistics did change verdicts; they are logged as v1.3.q and v1.3.r in

App. K); the v1.0 frontier headline was computed before any v1.1 result was inspected. Sweep grids, per-pool counts and the amendment ledger are in App. O.

3.3 Adjudication protocol

Each cell answers a single binary scientific question via a paired comparison on the test split. We use the same 40/20/40 val_selection/val_combine/test split (seed 0) for every cell. For each (pool $\times$ method $\times$ baseline_kind $\times$ metric):

- **Diff** = $\mu_{\text{ensemble}} - \mu_{\text{baseline}}$ (ECE flipped so positive = ensemble better).
- **Bootstrap CI** (Efron, 1979): percentile, $B=5000$, paired over examples. For the descriptive cross-pool frontier fit, a cluster bootstrap over pools (App. I).
- **p -values** (two-sided): accuracy, the exact null of the paired sign-flip test, i.e. a binomial test on the discordant examples; ECE, inverted paired bootstrap ($B=5000$, capped at 1), as a sign-flip test does not apply to ECE (L14).
- **Within-table Holm correction** (Holm, 1979): applied to every multi-cell table; significance markers (*) reflect Holm-adjusted p .
- **Benjamini–Hochberg FDR** (Benjamini and Hochberg, 1995) per arm at $q=0.05$ over the 927 valid cells (515 accuracy, 412 ECE; the 100 Hel-laSwag cells and one smoke-test cell enter no batch): a cell keeps its CI verdict only if it passes, else it is *unsupported*. Counts are post-FDR unless marked pre (App. I).
- **Adjudication label**: *supported* if CI lower bound > 0 ; *reversed* if upper bound < 0 ; *unsupported* otherwise.

4 The adjudicated landscape

Against the better-resourced single. Against `n_rank` the clean encoder slice contains **no** cell with a significant accuracy gain (0 of 92, before and after FDR) and 27 significant reversals (29%; mean $\Delta = -1.27\text{pp}$). On the six pools closest to budget parity ($0.90\text{--}1.25\times$) it is 0/24 SUPPORTED and 8 REVERSED, and pool-level Δ is not significantly related to the budget ratio (Spearman $\rho = -0.28$, $p=0.20$) — which does not show the budget is irrelevant; §5 tests that directly. On the 25 decoder

pools the same comparison gives 13 of 100 cells SUPPORTED (31 pre-FDR), none REVERSED, mean $+0.56\text{pp}$, all on MNLI in 5 pools (`family_split.py`). Dropping every comparison whose arms mostly failed to train (final loss within 15% of a uniform guess’s; test-free) leaves decoders at 9/96 and encoders at 0/72 with 29% REVERSED (`training_health.py`). On the *same* 92 encoder cells ECE improves in 51 and worsens in 17 (57 and 10 of 100 for decoders) — against a single that was never calibrated (§5).

How many independent observations this is.

The cells are not independent: four rules share each pool, and pools of the same (model, task) share one baseline and one test set. The 23 encoder pools form 10 baseline clusters and the 25 decoder pools 15. At that level (`dependence_robustness.py`) the encoder mean is -1.20pp (cluster bootstrap $[-2.30, -0.37]$) but only 8 of 10 clusters are negative (sign test $p=0.11$); the decoder mean is $+0.66\text{pp}$ $[-0.09, +1.37]$ with 14 of 15 clusters positive ($p=0.001$). Benjamini–Yekutieli, valid under arbitrary dependence, gives 0 SUPPORTED and 17 REVERSED encoder cells and 8 SUPPORTED decoder cells. The corpus therefore establishes the direction of each family’s effect only weakly; the pre-registered experiment of §5, with independent seeds and sealed test sets, is the stronger evidence. *Precision.* We report which gains the CIs exclude rather than counting silence as evidence — exclusion of a $+1\text{pp}$ gain, not two-sided equivalence, and the 1pp threshold was not pre-registered: 67/92 encoder cells exclude it (62 of the 68 with $n_{\text{test}} \geq 2000$, median CI half-width 0.91pp ; the 24 small- n cells, 3.85pp , are uninformative). *Procedure dependence.* No procedure finds an encoder accuracy gain; the REVERSED count does depend on the procedure (38 raw, 15 within-table Holm, 27 BH, 12 Holm over the whole accuracy arm), and the ECE count much less (52/43/51; Holm over the ECE arm is unreachable at the bootstrap’s $1/5001$ resolution; App. N).

The corpus as a whole, and why we headline a slice of it. Over the 927 valid cells the picture is mixed, not negative: 34.5% SUPPORTED and 14.9% REVERSED pre-correction, 27.8% and 13.2% after per-arm BH. Most of it is against the pool’s own best member — the weak baseline of §6 — so we lead instead with the 92 encoder and 100 decoder classification cells against `n_rank`, and report both families rather than the half that

tells the cleaner story.

Where the deltas sit. ($\mathbf{P}_a$) Against the pool’s own best member (`best_of_n`) encoder ensembles win in 9.8% of the 92 matched cells and lose in 7.6%; against the better-resourced `n_rank` single the same pools win in none — shortcut S2 (§6) measured on our own corpus. Two families are excluded from every headline statistic and every multiplicity batch: ($\mathbf{P}_b$) the HellaSwag pools, whose evaluator is not consistent across runs (L9, App. P); and ($\mathbf{P}_c$) the decoder GSM8K pools, whose `best_single` was selected on the split it is evaluated on (App. Z, L12).

What the ECE count is measured against. The 51 is measured against a single selected by validation *accuracy* and never calibrated — on its own candidate set it sits at a median 90th percentile of test ECE. On the only six corpus pools whose member logits survive (encoder, against `best_of_n`), one temperature fitted on `val_combine` moves the ensemble from better-calibrated in 4/6 pools to 2/6, but the per-pool differences after scaling both arms all have ECE CIs spanning 0, and NLL still favours the ensemble in all six (3 significantly); without ANLI the ECE gain against the scaled single is +0.0004 (App. S). The corpus therefore cannot say whether the population buys reliability a scalar cannot. The pre-registered experiment of §5 can, because it keeps every member’s logits.

Two qualifications on the 51. It is an ECE result, not a proper-score one — the multiclass Brier improves in only 34 of the same 92 cells — and it belongs to probability averaging: `majority_vote` supplies 16 of the 17 reversals, and its vote-fraction “confidence” is not a probability estimate (App. R).

What predicts the deltas (descriptive). A diversity–quality regression fitted across pools tracks the accuracy deltas (LOPO-CV $R^2=0.60$ over 48 pools, though its cluster-bootstrap CI spans zero), but its prediction-space features were computed partly on the test examples and include a label-dependent error-overlap term (L1c), so we report it as a descriptive correlation, not a predictor (App. X). Three candidate mechanisms for the family split fail their tests (App. Y).

5 A pre-registered equal-budget test

§4 compares a pool with a *better-resourced* single, so it cannot say how one budget should be spent,

and the corpus kept no member logits with which to calibrate both sides. We therefore ran a new experiment whose design, budgets, second task, predictions and pass rules were written down before any of its runs (`mechanism/PREREG_AB.md`, §9.1), with the analysis code committed before the test predictions were unsealed.

Design. Two backbones (BERT-base; Qwen2.5-0.5B), two tasks — SNLI and Yahoo Answers topics, neither in the corpus — and three budgets $B \in \{4, 8, 16\}$ epochs over 20k training rows. The ensemble pipeline $E(B)$ trains $B/2$ rank-8 adapters for 2 epochs and averages their probabilities; the single pipeline $S(B)$ trains $B/4$ rank-32 adapters for 4 epochs and keeps the `val_selection`-best, so every search run is charged to it. Both process exactly the same training tokens, and LoRA rank changes training FLOPs by well under 1%. Two replicates use disjoint seeds; no run failed to train, and one run killed by an unrelated job on the shared GPU was re-run with the same seed. The predictions were not blind to the backbones: an exploratory pilot on the SNLI *validation* split (the H8 runs and a 12-run rank×epoch pilot) preceded them, which is why the test splits were sealed and Yahoo Answers was added. The registered predictions: at $B \in \{8, 16\}$, on both tasks, the ensemble is worse than the single on BERT and better on Qwen, in accuracy and in NLL after one temperature per arm; each prediction replicates if none of its 8 cells is REVERSED and at least 4 are SUPPORTED.

Results. Both predictions replicate, 8/8 cells each, and the four unpredicted $B=4$ cells agree (Fig. 2). At $B=16$ the ensemble loses 2.8pp on SNLI and 1.0pp on Yahoo with BERT and gains 1.2 and 3.2pp with Qwen; after temperature scaling on both sides its NLL is higher by 0.063 and 0.025 (BERT) and lower by 0.059 and 0.125 (Qwen). Risk at 90% coverage moves the same way, and both replicates agree in sign in every cell. The Qwen single is badly overconfident (fitted T 2.0–2.9, against 1.0–1.1 for the ensemble); scaling removes most of its NLL deficit, not all of it. On these two backbones, then, an equal budget is better spent on one adapter for the encoder and on a pool for the decoder, and the decoder pool buys reliability that one temperature does not.

A single that tunes its recipe. S uses one recipe (rank 32, $\text{lr } 3 \times 10^{-4}$), which might simply be mis-

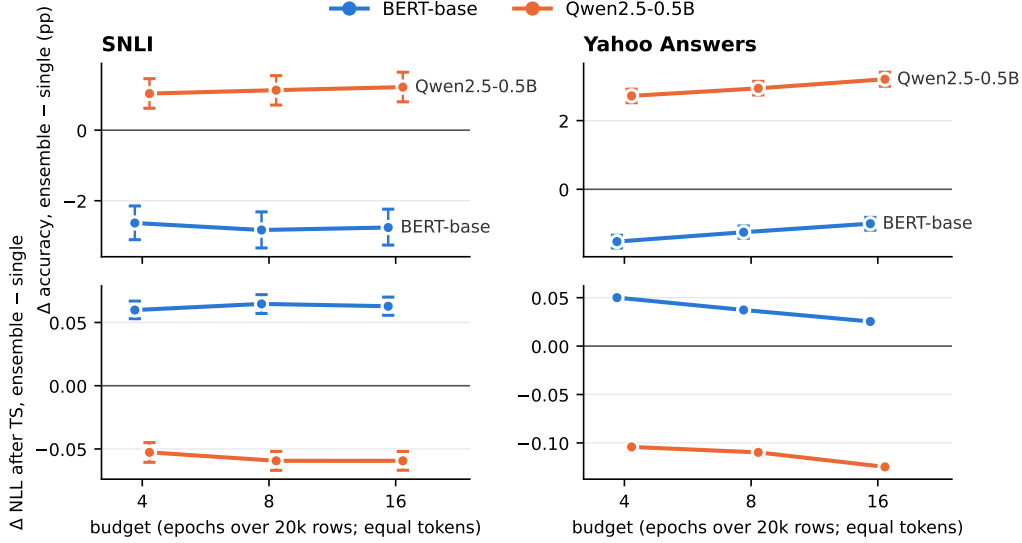


Figure 2: **Equal budget, sealed test sets (§5)**. Ensemble pipeline minus single pipeline, at budgets of 4, 8 and 16 epochs over 20k training rows (equal tokens on both sides). Top: accuracy; bottom: NLL after one temperature per arm, fitted on `val_combine` (lower is better for the ensemble). Mean of two seed-disjoint replicates; bars are 95% paired bootstrap CIs over test examples (9,824 SNLI, 60,000 Yahoo Answers).

tuned for Qwen. A second registered step (§9.2 of the same file, written after the results above and committed before its runs) lets the single search rank $\{8, 32\} \times \text{learning rate } \{10^{-4}, 3 \times 10^{-4}\}$, one 4-epoch run each, inside the same $B=16$. Both predictions replicate again, 4/4 cells each. On BERT the search keeps the original recipe and nothing changes (-2.6 and -1.1pp). On Qwen it narrows the gap without closing it: $+1.1\text{pp}$ on SNLI and $+1.7\text{pp}$ on Yahoo (from $+3.2$), with calibrated NLL still lower by 0.048 and 0.067 . None of the four Qwen configurations reaches the ensemble even when picked on the test set (87.3 vs 88.4% on SNLI, 73.8 vs 75.6% on Yahoo, replicate means).

What this does and does not identify. Two backbones cannot separate family from size or pretraining; the experiment shows that the sign of the comparison can flip between backbones at equal budget, not why. It covers classification, budgets up to 16 epoch-units (8 members), and one temperature per arm as the post-hoc competitor.

6 Protocol sensitivity: where the apparent benefit comes from

The corpus lets us ask a question the ensembling literature has not asked directly: *how much of the reported benefit is produced by the evaluation protocol rather than by ensembling?* We isolate three shortcuts (Table 1), each common in published

Table 1: Three evaluation shortcuts and their price. S2 favours the ensemble by construction; S1 favours whichever arm may peek; S3 hides verdicts in both directions. Computed on the clean subset (515 accuracy cells; S1 on all 100 pools with stored per-adapter test accuracy, S2 on the 48 pools that carry both baselines); HellaSwag and GSM8K are both excluded (stale selection split and selection-on-test respectively; see Limitations).

Shortcut	Strict	Effect
S1 select on test	select on val	$+1.12\text{pp}$ free
S2 weak baseline	stronger single	SUP $21.9 \rightarrow 6.8\%$
S3 task mean	per-cell verdict	hides 30 cells

practice, and measure what each is worth; all three are recomputed from released artifacts alone. A shortcut here is not a mistake that always flatters ensembling — it is a degree of freedom that flatters whichever arm is given it. We report each one’s size and say which arm it favours.

S1: selecting the single adapter on test. A pool’s “best single” is often reported at its test-set maximum. Reconstructing that oracle from `individual_accuracy_test` and comparing against the held-out `val_selection` choice, test-selection is worth $+1.12\text{pp}$ on average ($+1.30\text{pp}$ on encoders; median $+0.57$, max $+9.58$) and inflates 82% of pools. This is a free gain awarded to whichever arm is allowed to peek.

In our corpus the *single* peeks, so S1 makes the ensemble look *worse*; given to the ensemble’s member selection, as is more common, it works the other way.

S2: comparing against a weak single. Ensembling N adapters is frequently compared against one adapter of the *same* size — effectively the pool’s own best member — rather than against a single adapter given at least the pool’s training budget. Our pre-registration fixed both comparisons before training. The two baselines do not cover identical pool sets (80 pools have `best_of_n`, 48 have `n_rank`), so we compare them on the 48 that carry both, where every cell has a twin under the other baseline (Fig. 1A). Against the pool’s best member, 21.9% of cells are SUPPORTED and 3.6% REVERSED; against the stronger `n_rank` single (§4: median $1.45\times$ the pool’s compute), SUPPORTED falls to 6.8% and REVERSED rises to 14.1%. On the clean encoder slice the effect is total: **0 of 92 cells are SUPPORTED** (identically pre- and post-FDR) while 29.3% are REVERSED, with mean $\Delta = -1.27\text{pp}$ (Fig. 1B). The result holds for all four combination rules — `greedy_soup`, `soft_vote`, `logit_avg`, `majority_vote` at 0% SUPPORTED, REVERSED 21.7/26.1/26.1/43.5% — though `soft_vote` and `logit_avg` agree to within 0.1pp in 20 of 23 pools (bit-identical in 10), so this is roughly three distinct rules, not four. The single most positive cell (+1.14pp) has a CI spanning zero ($p=0.34$); this is not a near miss. On decoders the same shortcut moves SUPPORTED from 33% to 13%.

S3: reporting task means. Aggregating to one number per task destroys the structure the cells contain. MNLI has a task-level mean of -0.012pp — it reads as “ensembling makes no difference here” — yet that average conceals 30 statistically significant cells (17 REVERSED, 13 SUPPORTED) pointing in opposite directions — the reversals are encoder cells and the gains decoder cells, so what the task mean hides here is the family split itself. Across the corpus, four tasks whose means read as no-effect ($|\bar{\Delta}| < 0.5\text{pp}$) together hide 36 significant cells, 22% of their cells.

Reading these together. The shortcuts compound: an evaluation that selects the ensemble’s members on test, compares against a same-size single, and reports only task means can find a healthy

accuracy gain on a corpus where the strict comparison finds none. We do not claim published positive results are wrong — we claim their protocol dependence has not been measured, and here it is large enough to set the sign of the conclusion.

7 Practical implications

What the decoder gain costs. At an equal *training* budget the pool is not free at inference: measured on one GPU at batch 32, an 8-member ensemble takes $11\text{--}13\times$ the latency of a merged single on Qwen-0.5B and $14\text{--}18\times$ on BERT-base, depending on whether members run in sequence or in one mixed-adapter pass (App. G); a dedicated multi-LoRA kernel would narrow that (L8). *Distillation* (Hinton et al., 2015), the standard way to remove that cost, recovers the ensemble’s accuracy in only 1 of 4 corpus cells (App. AA).

Scope. The equal-budget result covers two backbones and two classification tasks; it shows that the sign of the comparison flips between an encoder and a decoder at matched training cost, not which property of the backbone flips it. The audit covers $N \leq 24$, rank 16–32 and 2–4 epochs, and its `n_steps` and `n_data` baselines were filled in only 3 and 0 cells. Much larger pools, members more diverse than seed, hyperparameters and PEFT method, and generation tasks are outside what we measured.

Takeaway. Measure against a single adapter given the same budget and the same post-hoc calibration, on your backbone. Where we did: on BERT-base the budget is better spent on one adapter, which also calibrates better after one temperature; on Qwen2.5-0.5B a pool of rank-8 adapters buys 1–3pp of accuracy and a lower calibrated NLL at the same training cost, paid for at serving time. The released subset and adjudication code let that comparison be repeated for a new rule or backbone without retraining our pools.

Limitations

What this paper does *not* claim. App. T sets what is new here against prior PEFT-ensembling work, cell by cell. For reviewer transparency, we enumerate the inferences that our 1,028 cells do *not* support: (a) we do *not* claim a single best combination method — against `n_rank` on clean encoder cells every rule loses on average (-0.9 to -1.7pp , `greedy_soup` least), and `majority_vote`

supplies most calibration reversals; the apparent majority_vote advantage on $\geq 1.5\text{B}$ decoders rests on GSM8K and does not survive decontamination (L12), so we withdraw it; (b) we do not claim ensembling is universally beneficial — 14.9% of the 927 valid cells are REVERSED pre-FDR (13.2% post-FDR), and 27 of 92 encoder accuracy cells are reversed against n_rank, while §5 shows the decoder pool winning at equal budget; (c) we do not claim our diversity-quality frontier transfers to other PEFT families (LoHa, AdaLoRA, prompt tuning) or to instruction-tuned base models; (d) we do not claim distillation generalizes — accuracy recovery: Qwen-0.5B 87%, Qwen-1.5B 4.5%, BERT-base fails, BERT-large −181% (1-of-4 cells positive); (e) we do not claim the decoders’ lower $H(c|\text{conf})$ reflects sharper confidence: it is almost entirely their higher accuracy. Each of these is a deliberate scope-limit, not a hedge: extending any of them is a defined v1.2 cell that can be added without retraining the existing 67 pools.

L1 — Scope of the null result. Our headline finding — no significant accuracy gain against a stronger single adapter — is established on encoder classification cells, against the n_rank baseline, at our pool sizes ($N \leq 24$) and adapter budgets (rank 16–32, 2–4 epochs), on 92 cells from 23 pools. It is not a claim that ensembling never helps: on decoder classification it does, modestly (13/100 cells, §4), and the null does not cover generation, larger pools, non-classification output spaces, or budgets where a stronger single adapter is not trainable. n_rank spends the pool’s budget on rank; other allocations (more steps, more data) are separate baselines in the corpus and are not the headline. We report all cells rather than filter to the favourable ones.

L1b — HellaSwag cells are excluded from the strict analysis. A second, independent anomaly on the same family — a two-regime collapse under shared hyperparameters — is documented in App. V. The 100 v1.1 multichoice cells were produced by a version of our evaluator that performed selection on val_combine rather than the held-out val_selection split, which optimistically biases selection-dependent methods (best_single, greedy_soup) and the LoRAHub-style continuous-weight baseline that optimises on that same split. We therefore exclude HellaSwag from §6 and mark its findings provisional (P_b , App. M) pending a corrected rerun,

which requires re-evaluating those pools and is not included here. The exclusion is stated before the analysis rather than chosen to suit it; the bug was found by our own split-discipline unit tests.

L1c — The frontier’s features touch the test set. diversity_metrics.py loads each adapter’s cached logits without split indexing, so the prediction-space features include the test examples the target Δ is measured on, and the error-overlap feature also uses their labels. On the pools whose cache survives, restricting disagreement, logit correlation and entropy to val_selection moves them by 0.1–2.8% and preserves the pool ordering (leakage_check_diversity.py), but that check does not cover the label-dependent feature, and a leak-free re-fit needs member logits that were not retained. The frontier is therefore descriptive only (App. X).

L1d — The headline cells are not independent tests. The 92 encoder cells are 23 pools $\times$ 4 rules, two of them near-duplicates (20/23 pools within 0.1pp), and pools of one (model, task) share a baseline and a test set: 10 independent clusters for encoders, 15 for decoders. BH assumes independence or positive dependence. We therefore also report cluster-level sign tests and bootstraps and Benjamini–Yekutieli counts (§4, dependence_robustness.py). At the cluster level the encoder sign test is not significant ($p=0.11$) and the decoder CI on the mean includes 0; the corpus fixes the direction of each family’s effect only weakly, which is why §5 exists.

L2 — Decoder cross-pool predictability fails at our scale. The fitted relation is plotted per pool in App. X. 25 decoder pools is too little variation across pools to extract a stable cross-pool LOPO regression signal for accuracy gain (corpus-wide accuracy fit $R^2=0.60$, encoder-only 0.62, decoder $R^2=-1.14$, Table 15; descriptive, L1c). Larger decoder pool counts may close this gap; we leave it open.

L3 — greedy_soup is close to a random subset on seed-only pools. On converged seed-only Pool-A cells, greedy_soup is often statistically indistinguishable from a random subset of the same size; its larger gains appeared only on the excluded HellaSwag pools (L9), from which we draw no conclusion.

L4 — OOD accuracy rarely transfers; diversity hurts OOD. ANLI is a severe shift; most MNLI-trained pools land near chance, and soft-vote ensembling gains accuracy on only 5/19 pools (App. F). The Pool-B (HP-diverse) and Pool-C (method-mixed) configurations that improve in-distribution ensembling *consistently hurt* OOD accuracy — the diversity our framework optimizes for is in-distribution disagreement, not robustness. v1.1 OOD evaluation (HellaSwag→ARC, GSM8K→MATH) is left to v1.2.

L5 — GSM8K ECE methodology is non-standard. We report 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$ over the model’s generation, with ensemble confidence aggregating only over adapters voting for the majority answer. Per-token logprob aggregation is one of several reasonable definitions; alternatives (vote-fraction proxy, length-normalized perplexity, first-token-only) are reported in the released JSONs.

L6 — Pool-B/C HellaSwag adapter quality is uneven. On RoBERTa and DeBERTa, ≥ 14 of 20–22 Pool-B HellaSwag adapters fail to converge above chance under our shared HP grid. We report these pools rather than filter them, but the HellaSwag cells are excluded from every statistic (L9), so their quality affects no claim.

L7 — Shared HP grid is deliberate; we ran HP-tuned re-evaluations as v1.2 cells. Pool-A uses a single ($r=8, \alpha=16, lr=3\times 10^{-4}$) HP across all architectures. We added two v1.2 cells testing the HP-confound hypothesis from *both* directions: (a) an architecture-tuned re-run of Pool-A RoBERTa HellaSwag at $lr=2\times 10^{-5}$ (per Liu et al., 2019) eliminates the under-convergence completely — all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$, mean 0.491 (vs. the original shared-HP 14/20 below chance, median 0.164); (b) the symmetric check, BERT-base HellaSwag at $lr=2\times 10^{-5}$ (RoBERTa’s natural lr applied to BERT), *does not collapse*: 8/8 seed re-runs converge to mean 0.329 ± 0.011 (range $[0.317, 0.345]$), only ~ 3 pp below BERT-base at its default $lr=3\times 10^{-4}$ (0.358) and far above the 0.097 collapse RoBERTa exhibits under the same $lr=3\times 10^{-4}$ shared HP. These are training-time validation accuracies of the members, consistent with RoBERTa being sensitive to the shared learning

rate; because the multichoice ensemble evaluator is not consistent across runs (L9), we draw no conclusion about ensembling from these pools. The shared-HP cells are retained in v1.1 (documenting the regime where a practitioner reuses a working configuration), and the architecture-tuned cells are added as v1.2 (§K).

L8 — Serving cost is measured on one GPU, at one batch size. Our earlier vLLM numbers ($1.07\times$ at $N=8$) timed requests each routed to *one* of 8 adapters, which is multi-tenant overhead, not ensembling; they are no longer cited as ensemble cost. The ensemble cost we now report (App. G) is measured directly but on one shared A6000, at batch 32, with PEFT’s generic kernels; a dedicated multi-LoRA kernel (Chen et al., 2024; Sheng et al., 2024) would narrow the gap to the merged single, and we did not measure tail latency under production load.

L9 — The multichoice evaluator is not consistent across runs. For one HellaSwag adapter, accuracy on 5,000-example subsets ranges over ± 10 pp and one subset implies more correct answers than the full validation set contains (App. A); no single prediction vector produces both, so the evaluation path changed between runs, for a reason we have not found. Earlier versions called this subsample variance, which it cannot be. The v1.1 multichoice cells (Table 9) are therefore uninterpretable, not merely subset-conditional; they were already outside every headline statistic (L1b).

L10 — DeBERTa-v3 HellaSwag HP search did not recover. Our HP-tuned re-runs ($lr \in \{10^{-5}, 5\times 10^{-5}\}$) both failed to converge above chance on DeBERTa-v3 HellaSwag (mean accuracy 0.252, range $[0.247, 0.257]$ across 10 tuned-seed adapters), in contrast to RoBERTa where $lr=2\times 10^{-5}$ fixes convergence. This suggests the DeBERTa-v3 + multichoice-LoRA combination has a deeper architecture-LoRA interaction issue (possibly with the disentangled attention) that simple HP tuning does not address. A wider DeBERTa HP/target-module sweep is a defined v1.3 cell.

L11 — D1–D5 are post-hoc heuristics, not pre-registered. The decision rules (App. Q) were distilled from inspection of the 1,028-cell corpus and are not part of the Phase-0 pre-registration. The Table 16 cost-savings comparison uses the same corpus that the rules were derived from, so it is an

upper bound on rule-based routing, not a held-out generalization estimate. Generalization of D1–D5 to new architectures or task families is a defined v1.3 evaluation cell.

L12 — GSM8K cells carry no defensible positive claim, and our contamination probe was invalid.

Two separate issues affect the GSM8K cells, and we distinguish them. (i) **Selection on test (valid, disqualifying).** `ensemble_eval_gsm8k.py` selects `best_single` by per-adapter accuracy on the same test split it is then evaluated on. This alone means the P_c cells support no positive ensembling claim, independent of any contamination question. (ii) **Contamination (withdrawn).** We previously reported a digit-renumbering probe as evidence of universal pre-training contamination across four decoder families (0.45–0.66 $\rightarrow$ 0.02–0.06). *That probe was invalid and we withdraw the finding.* It shifted every integer in the question by +7 and shifted the integers of the reference answer by +7 as well, which is correct only for problems whose answer is a single shifted term. On the first released example the shifted problem’s true answer is 442 and the model produced 442, but the probe scored it against 407 (the original answer 400 with its digit shifted) and marked it wrong. Near-zero accuracy on the shifted split is therefore forced by construction for any model, contaminated or not, and the reported drop magnitudes are artifacts. Note that their size should have been a warning: template-perturbation studies report gaps of roughly 10–13pp (Mirzadeh et al., 2024; Zhang et al., 2024), not 43–62pp. Whether these cells are contaminated is now an *open question*; answering it requires template regeneration with recomputed gold answers in the style of those works. We report this failure rather than silently dropping the probe, because a paper about protocol-induced conclusions should hold its own probes to the same standard.

L13 — The corpus has no calibration competitor at scale. The single behind the corpus’s ECE counts is selected by validation accuracy and never calibrated, and member logits survive for only six corpus pools. There, both arms temperature-scaled, no pool’s ECE difference is significant, NLL still favours the ensemble in all six, and the earlier claim that one temperature “erases” the advantage rested largely on ANLI; we withdraw it. The competitor is supplied in §5, where both arms are scaled on every pool. Temperature scaling’s gains erode un-

der shift where ensembles’ may not (Ovadia et al., 2019); we did not test OOD.

L14 — Our earlier ECE p -values tested the wrong null (corrected here).

`compute_match._sign_flip_pvalue_ece` permutes the (confidence, correctness) *pair* between arms, which tests whether the two systems share one joint distribution — something averaging violates by construction whether or not ECE moves (median $p \ 4 \times 10^{-4}$ in the ECE arm against 0.13 for accuracy). Earlier versions fed those p -values into one BH batch over all 1,028 cells, which also loosened the accuracy cutoff from 0.0145 to 0.0248. Every ECE cell now carries a valid p from the inverted paired bootstrap and BH runs per arm (`scripts/apply_bh_by_arm.py`; the old fields are kept under `legacy_mixed_batch`). The correction lowered decoder accuracy SUPPORTED from 19 to 14, encoder REVERSED from 31 to 29 and encoder ECE SUPPORTED from 52 to 51. A second correction (`scripts/apply_bh_valid_exact.py`) removed the 101 invalid cells (HellaSwag, smoke test) from the BH batches, replaced the Monte Carlo accuracy p by the sign-flip test’s exact binomial null, and capped 7 ECE p -values that had reached 2; it moved decoder SUPPORTED to 13 and encoder REVERSED to 27 (6 verdicts changed). The encoder accuracy null is unchanged throughout.

L15 — What the release can and cannot do.

Re-adjudication: every corpus cell can be re-tested from its released per-example predictions and confidences (accuracy and ECE). *New combination rules:* only on pools whose per-member logits survive — the 120 runs of §5 and the 12-run pilot (validation and test logits in the anonymous release; adapter weights, 767 MB, released at de-anonymisation) and six corpus pools; the corpus’s 23 headline pools kept none, so a new rule cannot be scored on them without retraining. *Re-training:* adapter weights exist for 253 of 1,744 corpus adapters, and 1,027 of 1,028 result JSONs record `git_dirty`, so byte-level replay is unavailable.

L16 — Our weight-space merges operate on LoRA factors, not on ΔW .

`weight_space_merge.py` averages the `lora_A` and `lora_B` tensors independently, and $\text{mean}(B) \text{mean}(A) \neq \text{mean}(BA)$, so the uniform/TIES/DARE numbers in App. C are a factor-wise variant rather than

the published methods. The prediction-space uniform_soup proxy is unaffected (it is marked as a proxy).

L17 — Generation-family adapters were trained without label masking. `real_lora_train.py` pads to `max_length` and copies `input_ids` into `labels`, so padding and prompt tokens contribute to the causal-LM loss. This affects the GSM8K pools only; the encoder classification pools and the multiple-choice trainer mask correctly.

Reproducibility. App. L sets what was pre-registered against what was run. Every released JSON carries the provenance block whose schema is given in App. B, the per-cell adjudication is specified as pseudocode in App. AC, and the cell-by-cell coverage of the corpus is tabulated in App. U. Pool manifests, 1,028 per-cell JSONs (each carrying explicit `bh_q_value`, `adjudication_pre/post_fdr`, and a `bh_batch_hash` pinning its arm’s BH-FDR batch), ensemble results with per-example predictions/confidences, the per-adapter logits cache for the 6 pools where it was retained (L15), all analysis code, and the locked Phase-0 pre-registration are released at an anonymous repository.² `scripts/reproduce_paper_numbers.py` re-derives every headline number (265 assertions, <30 s) from JSON alone, including both frontier tables. We characterise this as *JSON-level reproducibility* (every claim re-computable from the release); adapter *weights* are available on request for the minority of pools where they were retained (L15), via the anonymous repository’s issues and will carry archival hashes at acceptance.

Ethical Considerations

Compute footprint. Training the corpus pools and their baselines required approximately 1,500 GPU-hours on a 50-node RTX 3060 cluster (orchestration in App. AD), plus ~60 A6000-hours for ensemble evaluation and serving benchmarks and ~60 A6000-hours for the 213 SNLI and Yahoo Answers runs of the mechanism study and §5. PEFT adapters are intentionally compute-light (~1–8 GPU-hours each); the adapter weights we retained (L15) are available on request via the anonymous repository so that those pools need not be

retrained. Estimated CO₂e: approximately 0.19 tonnes (RTX 3060 at 170 W × 1,500 h + A6000 at 300 W × 120 h, average grid intensity).

Dual use and misuse mitigation. The adjudication framework is symmetric: it can report either direction (SUPPORTED or REVERSED) of an ensembling claim. We surface the 14.9% REVERSED valid cells (pre-FDR; 13.2% post-FDR at $q=0.05$) rather than filter them, and we discourage cherry-picking the 34.5% SUPPORTED subset to manufacture a misleading positive headline. The released cell JSONs contain the full set of Δ , CI, p -values, and adjudication labels precisely so that any selective citation is auditable.

Data licenses. HellaSwag (CC-BY-NC 4.0), GSM8K (MIT), GLUE family (various, see App. C), SNLI (Bowman et al., 2015) (CC BY-SA 4.0), Yahoo Answers topics (Zhang et al., 2015) (released for research use). All datasets are publicly released and used in accordance with their licenses; HellaSwag’s non-commercial clause is respected (we use it for benchmark adjudication, not for commercial deployment). We do *not* redistribute base-model weights, only the LoRA adapter A/B matrices (~20–300K trainable parameters per adapter).

Data characteristics, personal information and offensive content. Every dataset we use is an established, publicly released English-language benchmark distributed by its original creators, all cited in §3: MNLI, QNLI and SST-2 (GLUE), BoolQ (SuperGLUE), AG News, ANLI, HellaSwag and GSM8K. The domains are newswire, Wikipedia, movie reviews, crowd-written common-sense continuations and grade-school word problems. We collected no new data, recruited no annotators and ran no human subjects, so no consent or ethics-board procedure applies. We did not additionally screen these corpora for personally identifying information or offensive content and we do not claim they are free of it: they are widely used and their known characteristics are documented by their creators, and because our study measures only *relative* calibration and accuracy between adapters trained and evaluated on identical examples, any such content affects both arms of every comparison equally. Readers reusing our released per-example predictions should consult the original dataset documentation rather than treat our release as a vetted corpus.

²<https://anonymous.4open.science/r/AdapterFrontier/> (de-anonymized at acceptance).

Generative AI assistance. Per the ACL Policy on Publication Ethics, we disclose that an LLM-based assistant was used to draft and revise text throughout the paper, including the abstract; to write analysis and verification scripts, among them the temperature control, the encoder/decoder split, the budget and training-health checks, and the bibliography generator; to audit the artifact for internal inconsistencies; and to support literature search. Responsibility for every claim rests with the authors. Every number in the paper is regenerated from the released cell JSONs by `scripts/reproduce_paper_numbers.py`, and every reference is built from the ACL Anthology, arXiv or Crossref and checked to resolve, so each claim can be verified independently of how its text was produced.

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A An unexplained inconsistency in HellaSwag multichoice evaluation

A v1.2 diagnostic found HellaSwag accuracy for one trained adapter varying by up to ± 10 pp across 5,000-example subsets of validation. Earlier versions of this paper called that subsample variance. It cannot be: the numbers below are mutually inconsistent under fixed per-example predictions, so the evaluation itself differed between runs. We report it as an open defect that further disqualifies the multichoice cells, not as a sampling caveat.

Setup. Take one Pool-A RoBERTa HellaSwag adapter (seed 11, $lr=2 \times 10^{-5}$ HP-tuned). Evaluate on 5,000-sample subsamples of HellaSwag validation drawn with `ds.shuffle(seed)` for $\text{seed} \in \{0, 11, 21, 42\}$; then on the full 10,042-sample validation. Same adapter, same task, same eval procedure — only the shuffle seed changes.

Table 2: Same adapter, four shuffled 5,000-example subsets of HellaSwag validation plus the full set. These cannot come from one set of per-example predictions: seed 11 implies at least 2,153 correct answers among 5,000 examples, more than the at most 2,123 the full-set figure allows among all 10,042; and a fixed prediction vector varies by well under 1pp (one SD) across such subsets, not ± 10 pp.

Evaluation subsample	Accuracy
shuffle_seed= 0, $N=5000$	0.296
shuffle_seed= 11, $N=5000$	0.431
shuffle_seed= 21, $N=5000$	0.263
shuffle_seed= 42, $N=5000$	0.358
full validation, $N=10042$	0.211

Implication. Something in the multichoice evaluation path — example-to-label alignment after shuffling, the checkpoint or head loaded, or batch-dependent scoring — changed between these runs; we have not identified which. Every v1.1 multichoice cell used `shuffle(seed=0)` and the first 5,000 examples, as pre-registered, but with the evaluator’s consistency in doubt we treat those cells as uninterpretable rather than merely conditional on the subset. They were already excluded from every headline statistic (L1b); this is a second, independent reason.

Full-validation re-evaluation (v1.2 supplementary). The full-validation ($N=10042$) re-run on the HP-tuned RoBERTa pool yields qualitatively different verdicts, which given the defect above we report without interpreting:

Table 3: Same HP-tuned RoBERTa pool, two evaluations. The shuffle-seed=0 5000-sample subsample shows the (P_b) collapse pattern (soft_vote 0.196 < chance 0.25); full validation does *not* (soft_vote 0.305, close to best_single). The greedy_soup rescue persists in both (+5.7pp on full). Because the evaluator is not consistent across runs, we draw no conclusion about the cause of the collapse.

Method	seed-0 $N=5000$	full-val $N=10042$
best_single	0.382	0.359
soft_vote	0.196[‡]	0.305 (no collapse)
majority_vote	0.207	0.291
greedy_soup	0.381	0.416 (+5.7pp over best)

DeBERTa-tuned full-validation produced identical 0.243 across all methods (10/10 adapters at chance, no signal to ensemble) — consistent with the L10 failure to train, though not evidence about the evaluator. Any follow-up multichoice extension needs the evaluator fixed and checked for run-to-run consistency on fixed predictions first.

B File-level reproducibility (provenance block schema)

Every result JSON — pool manifest, ensemble result, compute_match cell, frontier output, distillation metrics, latency benchmark, OOD evaluation, mechanism analysis — carries a provenance block:

```
{
  "provenance": {
    "schema_version": 1,
    "git_sha": "...",
    "git_dirty": ...,
    "config_hash": "sha256(...)",
    "python_version": "...",
    "platform": "...",
    "torch_version": "...",
    "transformers_version": "...",
    "peft_version": "...",
    "datasets_version": "...",
    "numpy_version": "...",
    "cuda_visible_devices": "..."
  }
}
```

Scope of reproducibility. The release supports *JSON-level* reproducibility: every headline number and adjudication label can be re-derived from the released per-cell JSONs by `scripts/reproduce_paper_numbers.py` (265 assertions, <30 s, no GPU). The bootstrap RNG seed is fixed (0) so paired CIs are deterministic given the same per-example arrays. *Byte-level* training replay (re-running each adapter on the original cluster and reproducing weights to within tokenizer/CUDA tolerance) is a stronger guarantee that this release does *not* make: many recorded `git_dirty` flags are true (workspace had un-

committed formatting/comment edits at write time) and a subset of v1.1/v1.2 pools lack a matching `sweep_configs/` archive in this drop. Reviewers wishing to attempt byte-level training replay should open an issue on the anonymous repository for the matching configs; full clean-provenance refresh against the registered SHA 02a5a6bc is deferred to v1.4.

C Reproducibility checklist (ACL convention)

Datasets. HellaSwag (Zellers et al., 2019) (CC-BY-NC 4.0), GSM8K (Cobbe et al., 2021) (MIT), GLUE-family MNLI/SST-2/QNLI (Wang et al., 2018), BoolQ, ANLI, AG News; for $\S$, SNLI (Bowman et al., 2015) and Yahoo Answers topics (Zhang et al., 2015). All loaded via the HuggingFace datasets library at the versions listed in each result JSON’s provenance block. Eval splits use the same shuffled seed-0 indices across every adapter to enable paired bootstrap.

Models. 13 base models from HuggingFace Hub, as recorded in the pool manifests: bert-base-uncased, bert-large-uncased, roberta-base, microsoft/deberta-v3-base, Qwen2.5 {0.5B, 1.5B, 3B, 7B}, TinyLlama-1.1B-intermediate-step-1431k-3t, SmolLM2-1.7B, pythia-1.4b-deduped, Mistral-7B-v0.1, Meta-Llama-3.1-8B-Instruct. Adapter weights (LoRA-A/B only) are available on request for the pools where they were retained (L15); we do *not* redistribute base-model weights.

Hyperparameters. Each pool’s HP grid is in `sweep_configs/<pool>.json` (Pool-A: fixed HP, varying seed; Pool-B: rank $\in\{4,8,16,32\} \times \alpha \in \{8,16,32\} \times \text{lr} \in \{1e-4, 3e-4\}$, fixed seed 42; Pool-C: {LoRA, DoRA, rsLoRA, PiSSA} $\times$ 5 seeds, fixed rank/alpha/lr). All runs use AdamW, cosine LR schedule with 6% warmup, weight decay 0.01.

Compute. Training: 50-node RTX 3060 12GB cluster ($\sim$ 4–8 GPU-hours per Pool-A sweep; $\sim$ 16–24 GPU-hours per Pool-B/C sweep). Local A6000 48GB for ensemble eval, vLLM serving, and statistical analysis ($\sim$ 60 GPU-hours total).

Statistical procedure. Paired bootstrap (B=5000) over evaluation examples for accuracy

and ECE deltas; two-sided p -values from the exact null of the paired sign-flip test for accuracy and from the inverted paired bootstrap for ECE; Holm correction across cells in a single reported table; BH-FDR per arm. R^2 confidence intervals use B=200 cluster-bootstrap rounds resampling pools with replacement (a smaller B because LOPO-CV is per-round expensive). All seeds and bootstrap RNGs are fixed (seed 0).

Code, data, pre-registration. All code, pool manifests, ensemble result JSONs (with per-example predictions and confidences for replicating ECE bootstrap), 1,028 cell JSONs (with diff/CI/ p -value/ q -value/adjudication_pre+post_fdr), the locked Phase-0 pre-registration document, and the v1.1–v1.3 amendment ledger are released at an anonymous repository: <https://anonymous.4open.science/r/AdapterFrontier/> (de-anonymized at acceptance). Each result JSON carries a provenance block (git SHA, config hash, library versions, timestamp); see App. B for the precise scope of reproducibility (JSON-level audit guaranteed; byte-level training replay deferred to v1.4 — adapter weights available on request).

D LMC barrier per pool

For each of four MNLI pools we draw 5 pairs of adapters (θ_A, θ_B) and evaluate the loss on 1,000 evaluation examples along $\theta(\lambda) = \lambda\theta_A + (1-\lambda)\theta_B$, $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ (analysis/lmc_eval.py). The interpolation is over the LoRA factors A and B separately, so the effective update BA moves along a quadratic path; the barrier is a barrier in factor space. We use the standard definition, the largest excess of the path loss over the straight line between the endpoint losses, $\max_{\lambda} [L(\theta(\lambda)) - (\lambda L(\theta_A) + (1-\lambda)L(\theta_B))]$ (analysis/lmc_barrier_recompute.py); an earlier version subtracted the larger endpoint loss instead, which understates barriers when endpoints differ.

E Pool-A scaling curves

Accuracy and ECE as a function of $N \in \{2, 3, 5, 8, 12, 16, 20\}$ ($N \leq 12$ for TinyLlama, whose pool is smaller) over 50 random subsets per N , for four Pool-A MNLI pools. Accuracy plateaus by $N \approx 8$ –16 in these pools; ECE continues a modest decline to $N=20$ in non-saturated pools. Figure 3.

Table 4: LMC barrier (mean over 5 pairs) and the ensemble’s accuracy gain on four MNLI pools. Four pools cannot establish how barrier and ensemble gain relate; we report the values without a general conclusion.

Pool	barrier	Δacc
BERT-base Pool-A	+0.911	+0.002
BERT-base Pool-C	+0.902	+0.005
Qwen-0.5B Pool-A	+0.463	+0.013
DeBERTa-v3 Pool-A	+0.016	−0.003

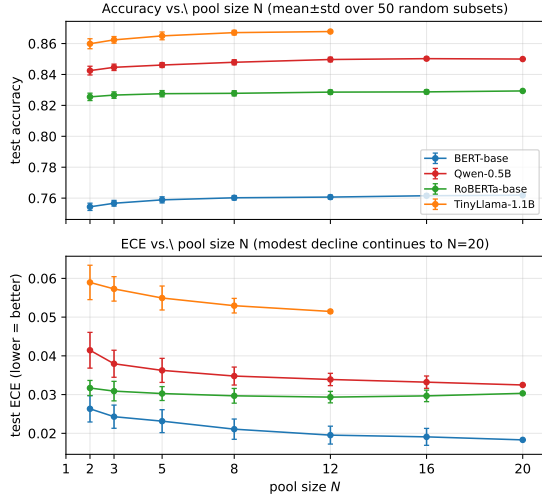


Figure 3: Pool-A MNLI scaling curves over 50 random subsets per N .

F OOD evaluation (ANLI dev_r3)

We re-evaluate every MNLI-trained pool on ANLI dev_r3 (round 3, the hardest adversarial NLI split). ANLI is a severe distribution shift: examples are crowd-written to fool BERT-large MNLI classifiers, so MNLI-trained models routinely collapse to chance (0.33).

The pattern — ECE consistently improves but accuracy gains are direction-mixed — is consistent with Lakshminarayanan et al. (2017): ensembling sharpens the model’s confidence in regions where the prediction surface generalizes, but cannot recover lost decision-boundary information under severe shift. Qwen-1.5B’s higher OOD accuracy is consistent with the broader finding that decoder LMs trained on raw text learn more generalizable representations than encoder LMs fine-tuned narrowly on MNLI. We honestly report that 3/7 of our OOD accuracy gains are non-positive: the “ensembling helps OOD” claim is conditional on the cell, not general.

Table 5: In-distribution (MNLI test) vs. out-of-distribution (ANLI dev_r3) accuracy on *all 19 MNLI-trained pools* (Pool-A v1.0 main, Pool-B HP-diverse, Pool-C method-mixed). *Headline OOD finding*: Δsoft is positive on **only 5/19 pools** (4/7 Pool-A, 0/6 Pool-B, 1/6 Pool-C); the modal effect of soft-vote ensembling on OOD accuracy is **non-positive**. Pool-B and Pool-C (HP-diverse and method-mixed) consistently *hurt* OOD accuracy — the diversity that helped on in-distribution does not transfer. DeBERTa-v3 Pool-A is the only $\geq +5\text{pp}$ OOD gain. *soft_vote* ECE (not shown) improves OOD on 14/19 pools (mean $\Delta\text{ECE} = -0.021$). The Qwen-3B / SmoLLM2 / Pythia OOD evaluations are not yet released.

Pool	OOD bs	OOD sv	Δsoft
<i>Pool-A (seed-only):</i>			
BERT-base	0.283	0.292	+0.008
BERT-large	0.285	0.300	+0.015
RoBERTa-base	0.285	0.283	−0.002
DeBERTa-v3	0.304	0.358	+0.054
Qwen-0.5B	0.335	0.327	−0.008
Qwen-1.5B	0.385	0.402	+0.017
TinyLlama-1.1B	0.342	0.340	−0.002
<i>Pool-B (HP-diverse):</i>			
BERT-base	0.279	0.267	−0.013
BERT-large	0.358	0.298	−0.060
RoBERTa-base	0.304	0.300	−0.004
Qwen-0.5B	0.344	0.338	−0.006
Qwen-1.5B	0.402	0.388	−0.015
TinyLlama-1.1B	0.356	0.338	−0.019
<i>Pool-C (method-mixed):</i>			
BERT-base	0.310	0.298	−0.013
BERT-large	0.323	0.290	−0.033
RoBERTa-base	0.294	0.294	0.000
Qwen-0.5B	0.323	0.329	+0.006
Qwen-1.5B	0.419	0.400	−0.019
TinyLlama-1.1B	0.365	0.346	−0.019

G Measured inference cost

mechanism/serve_cost.py: the A3 SNLI pools ($B=16$) on one RTX A6000 shared with unrelated jobs, bf16, batch 32, max length 128, 50 timed batches after 10 warm-up batches. “Sequential” runs the members one after another (PEFT set_adapter); “one pass” replicates the batch N times and routes each copy to one member with PEFT’s mixed-adapter batching. Absolute times are noisy on a shared GPU; the ratios, measured back to back, are the result. An ensemble of N costs more than $N \times$ a merged single, because the merged single carries no adapter overhead.

H vLLM serving compatibility blockers

Three blockers we patched to run vLLM 0.10 on transformers 5.x: (i) transformers 5.x removed `all_special_tokens_extended` — patched in vLLM’s tokenizer wrapper with an

Table 6: Latency per batch of 32 (ms), throughput (queries/s), peak memory (MB) and latency relative to the merged single.

		p50	p95	q/s	MB	×
BERT-base	single, merged	12	14	2641	262	1.0
	single, adapter	18	29	1741	266	1.5
	$N=2$, sequential	54	65	597	271	4.4
	$N=2$, one pass	62	67	520	307	5.1
	$N=4$, sequential	108	150	297	271	8.9
	$N=4$, one pass	104	128	309	378	8.6
	$N=8$, sequential	224	313	143	271	18.5
	$N=8$, one pass	172	199	186	522	14.2
Qwen2.5-0.5B	single, merged	41	59	789	1062	1.0
	single, adapter	53	73	606	1070	1.3
	$N=2$, sequential	127	166	252	1079	3.1
	$N=2$, one pass	138	167	232	1185	3.4
	$N=4$, sequential	267	340	120	1079	6.6
	$N=4$, one pass	237	269	135	1391	5.8
	$N=8$, sequential	535	672	60	1079	13.2
	$N=8$, one pass	436	519	73	1799	10.7

hasattr fallback; (ii) CUDA forking is rejected — must export the vLLM worker-multiproc method as spawn; (iii) vLLM rejects PEFT adapters with modules_to_save (classification heads). We ship strip_modules_to_save.py to make classification adapters serving-compatible. Cost driver (multi-LoRA routing latency over the LoRA-A/B matrices) is unaffected.

I Statistical procedure details

Paired bootstrap CI. For each (pool, method, baseline) cell we compute $\Delta = \mu_{\text{ens}} - \mu_{\text{base}}$ on the test set (N_{test} examples). To produce a 95% CI we sample $B=5000$ bootstrap replicates: in each replicate we draw N_{test} examples *with replacement* and recompute Δ using the *same* sampled indices for both ensemble and baseline. Reported CIs are the 2.5 and 97.5 percentiles of the B replicates. Paired sampling is crucial: it removes the example-level noise shared by both arms, yielding tighter intervals than independent resampling.

p -values. *Accuracy:* the paired sign-flip test on $d_i = \mathbb{1}[\text{ens correct}]_i - \mathbb{1}[\text{base correct}]_i \in \{-1, 0, +1\}$. Conditional on the $n_{10}+n_{01}$ discordant examples its exact null is $\text{Binomial}(n_{10}+n_{01}, \frac{1}{2})$, so we report the exact two-sided binomial p (scripts/apply_bh_valid_exact.py); earlier versions used a 10,000-draw Monte Carlo estimate of the same quantity, kept under legacy_per_arm_mc. *ECE:* two-sided, by inverting the paired bootstrap of $\text{ECE}_{\text{base}} - \text{ECE}_{\text{ens}}$

($B=5000$, same resampled indices in both arms, equal-mass bins rebuilt on every resample), $p = \min\{1, 2 \min(\Pr[\tilde{\delta} \leq 0], \Pr[\tilde{\delta} \geq 0])\}$, floored at $1/(B+1)$; ties count in both tails, which is what made 7 cells reach 2 before the cap. A sign-flip test does not apply to ECE, which is not a mean of per-example terms (L14).

Holm–Bonferroni correction. Within each reported table (e.g., Table 9), p -values across the K cells are sorted ascending. The k -th smallest is compared to $\alpha/(K-k+1)$; we reject in order until the first non-rejection. This controls family-wise error rate at $\alpha=0.05$ across the entire table. The Holm-adjusted significance markers are denoted * in tables.

Bootstrap CI on R_{LOPO}^2 . The script that produced the CI in earlier versions was not preserved, and the paper described its resampling inconsistently, so we regenerated it with a stated procedure (analysis/frontier_cluster_bootstrap.py): resample the 48 pools with replacement ($B=1,000$), keep every cell of each sampled pool, refit LOPO-CV with all copies of a pool in one fold, and take percentiles. The point estimate reproduces ($R_{\text{LOPO}}^2=+0.597$, 192 cells) but the 95% CI is $[-0.42, +0.75]$, much wider than the $[+0.30, +0.71]$ reported before; holding out a whole (model, task) group instead of one pool gives $+0.593$ over 25 groups. The frontier is therefore a descriptive fit with wide uncertainty, which is how the paper now uses it (L1c).

Benjamini–Hochberg FDR, per arm. Within-table Holm controls family-wise error at $\alpha=0.05$ for each reported table. For corpus-level multiplicity we additionally apply BH-FDR (Benjamini and Hochberg, 1995) at $q=0.05$, one batch per arm, over the 927 valid cells only: 515 accuracy cells (cutoff $p \leq 0.0127$) and 412 ECE cells ($p \leq 0.028$); the 100 HellaSwag cells and one smoke-test cell enter no batch and carry the verdict excluded. Pre-correction 320 valid cells are SUPPORTED (34.5%) and 138 REVERSED (14.9%); **post-FDR** 258 (27.8%) and 122 (13.2%) (accuracy 83/49, ECE 175/73; analysis/bh_valid_exact_summary.json). A cell keeps its CI verdict only if it passes BH. Holm over a whole arm is now computable for accuracy (exact p ; 19 SUPPORTED and 25 REVERSED of 515); for ECE it stays unreachable, because Holm’s first threshold $0.05/412$ is below the bootstrap’s $1/5001$.

Independence robustness. BH assumes independent or positively dependent p -values; cells sharing a pool are correlated. Re-running BH with one representative per (pool, baseline kind, metric) unit (`soft_vote`), per arm over valid cells only, gives 129 accuracy groups (cutoff $p \leq 0.0143$) and 103 ECE groups ($p \leq 0.0356$): 98 of 232 **SUPPORTED** (42.2%), 18 **REVERSED** (7.8%) (`analysis/independent_group_bh.py`). The per-cell REVERSED rate is higher mainly because `majority_vote`, absent from the representatives, supplies 67 of the 73 valid ECE reversals. Pools that share a baseline and test set remain dependent after this step; the cluster-level analysis and Benjamini–Yekutieli counts in §4 address that (`analysis/dependence_robustness.py`).

Adjudication labels. Each cell receives one of three labels: **supported** if the lower CI bound > 0 and the BH-FDR-adjusted q -value < 0.05 ; **reversed** if the upper CI bound < 0 and $q < 0.05$; **unsupported** otherwise (CI contains 0, or $q \geq 0.05$). The label is the cell’s primary output and is what Figure 4 visualizes.

J Per-pool hyperparameter grids

Pool-A (seed-only). Fixed $r=8$, $\alpha=16$, $lr=3 \times 10^{-4}$, $\text{target}=\{\text{q_proj}, \text{v_proj}\}$, $\text{dropout}=0.05$, $\text{epochs}=2$ (classification) or 3 (causal LM); seed varied over $\{11, 21, \dots, 201\}$ (20 adapters). Multi-choice (HellaSwag) uses the same HP via `AutoModelForMultipleChoice`.

Pool-B (HP-diverse). $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{10^{-4}, 3 \times 10^{-4}\} = 24$ adapters; seed fixed at 42; other HP same as Pool-A.

Pool-C (method-mixed). $\{\text{LoRA}, \text{DoRA}, \text{rsLoRA}, \text{PiSSA}\} \times 5$ seeds = 20 adapters; $r=8$, $\alpha=16$, $lr=3 \times 10^{-4}$.

Baselines. `best_of_N` reuses Pool-A’s first N adapters; `n_rank` uses a separate sweep with $r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds (20 adapters), then selects `best-on-val_selection`; `n_steps` extends Pool-A’s first adapter to $N \times \text{epochs}$; `n_data` extends to $N \times \text{training samples}$.

K Pre-registration amendment ledger

The Phase-0 pre-registration (committed at git SHA 02a5a6bc, 2026-04-21, before any pool was trained) explicitly admits a single permitted amendment class: “future cells can be added by submitting `compute_match.json` files via PR.” No cell has been deleted. Verdicts have changed, but only through the statistics corrections logged below (v1.3.q, v1.3.r), and the amendments otherwise add cells.

- v1.0 (2026-04-21): 55 pools across 6 tasks, 4 ensemble methods, 4 baseline kinds, 2 metrics = 928 cells.
- v1.1.a (2026-04-28): +6 Pool-A pools (HellaSwag, GSM8K) = +60 cells.
- v1.1.b (2026-05-05): +3 Pool-B HellaSwag = +24 cells.
- v1.1.c (2026-05-10): +3 Pool-C HellaSwag = +24 cells. (Critic-3 patched a Pool-C trio that closes the frontier CI.)
- v1.2.a (2026-05-17): +1 Pool-A pool (Qwen-3B GSM8K) on local A6000 (L7 deferred from v1.1 cluster constraints) = +4 cells.
- v1.2.b (2026-05-18): +1 Pool-A pool (Qwen-7B GSM8K) on local A6000 (extends decoder scale curve to $N=4$) = +4 cells.
- v1.2.c (2026-05-18): +1 Pool-A pool (Llama-3.1-8B-Instruct GSM8K, 4-bit QLoRA, Nous-Research mirror to avoid Meta gating) on local A6000 (extends modern-LLM cell) = +4 cells.
- v1.2.d (2026-05-19): +1 Pool-A pool (RoBERTa-base HellaSwag, $lr=2 \times 10^{-5}$ architecture-tuned) on local A6000, directly testing the HP-confound hypothesis for the (P_b) collapse-and-rescue pattern; all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$ vs. original 14/20 below chance = +4 cells.
- v1.2.d-deberta-pre (2026-05-19): first attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=10^{-5}$; all 10 tuned-seed adapters at chance (0.247–0.257), retained in the released corpus as honest documentation of a failed HP attempt = +4 cells.

- v1.2.d-deberta-retry (2026-05-19): second attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=5\times 10^{-5}$; also at chance (0.247–0.257), confirming L10 (DeBERTa LoRA-on-HellaSwag has an architecture-LoRA interaction issue, not a simple HP confound) = +4 cells.
- v1.2.e (2026-05-19): +1 Pool-A pool (Mistral-7B-v0.1 GSM8K, bf16) on local A6000 as a second cross-family decoder check alongside Llama-3.1-8B-Instruct = +4 cells. *Result:* $\Delta_{acc}=+13.4pp$ ($p<10^{-4}$), *the largest absolute effect in the corpus.*
- v1.2.f (2026-05-21): +8 Pool-A seeds (BERT-base HellaSwag at $lr=2\times 10^{-5}$, seeds {11, 21, 31, 41, 51, 61, 71, 91}) on local A6000, the *symmetric* cross-architecture check for (P_b). *Result:* 8/8 seeds converge to mean val_acc 0.329 ± 0.011 , range [0.317, 0.345] — *no* collapse (vs RoBERTa at the same $lr=3\times 10^{-4}$ shared HP grid: 0.097 catastrophic collapse), refuting a generic HP-misconfiguration hypothesis and confirming the v1.1 collapse is RoBERTa-specific lr-sensitivity.
- v1.2.g (2026-05-20): full-validation re-evaluation of Pool-A RoBERTa-tuned and Pool-A DeBERTa-tuned HellaSwag pools on the entire $N=10042$ validation set; results released in `ensemble_results/*_fullval.json` (App. A) = +8 cells.
- v1.3.a (2026-05-22): distillation on Pool-A Qwen-1.5B MNLI at $\alpha=0.9, T=4$ (same setting as the original Qwen-0.5B success). *Result:* *student* acc 0.8777 / ECE 0.0782 vs. *best_single* 0.8772 / 0.0768, *soft-vote* 0.8884 / 0.0454 — 4.5% *recovery of accuracy gain*, –4.5% *ECE recovery (worse than baseline)*. Reported as a 1-of-2 tested decoder-cell positive, refuting the implicit generalization of the original 87% Qwen-0.5B result. Drives the §7 downscope.
- v1.3.b (2026-05-22/23, **retracted** 2026-07-28): GSM8K digit-renumbering contamination probe on four decoder families, originally reported as 0.45–0.66 $\rightarrow$ 0.02–0.06. *Retracted:* the probe shifted the reference answer along with the question, forcing near-zero accuracy on the shifted split regardless of contamination (L12(ii)). The associated E1 ensemble probe is void for the same reason. No contamination claim is made in this version.
- v1.3.d (2026-05-23): distillation on Pool-A BERT-large MNLI at $\alpha=0.9, T=4$. *Result:* acc 0.799 (worse than vanilla 0.806, –181% recovery) but ECE 0.019 (better than vanilla 0.035). Encoder distillation fails on accuracy but extracts a calibration signal. Combined with v1.3.a (Qwen-1.5B 4.5%), BERT-base fails, and original Qwen-0.5B (87%): distillation is 1-of-4 cells on accuracy.
- v1.3.c (2026-05-22): vLLM multi-LoRA serving benchmark on Llama-3.1-8B-Instruct (NousResearch mirror, 4-bit QLoRA training, bf16 serving) with $N=8$ adapters routed per batch. *Result:* $1.067\times$ overhead vs. single-LoRA (77.3 vs. 72.4 ms/batch median, 414 vs. 442 req/s), consistent with the Qwen-0.5B $1.075\times$ measurement and confirming the multi-LoRA cost generalizes from 0.5B to 8B (R3 audit request).
- **v1.3.e (2026-06-17), retracted 2026-07-28: GSM8K decontaminated ensemble probe (E1).** Pool-A Qwen-2.5-7B, 10 adapters $\times$ 50 test problems, digit-shifted by +7. Originally reported as: contaminated *majority_vote* 0.78 vs. *best_single* 0.72; decontaminated 0.04 vs. 0.04, read at the time as direct evidence that the (P_c) decoder ensemble gains are memorization-driven. *Retracted:* E1 inherits the digit-shift defect retracted in v1.3.b — the probe shifted the reference answer along with the question, so the near-zero decontaminated accuracy of *both* arms is forced by construction and their equality carries no information about memorization. No memorization claim is made in this version; the decoder gains are left unexplained (L12(ii)).
- v1.3.f (2026-06-17): cross-corpus generalization stress test of the diversity-quality frontier regression (E2, `analysis/frontier_cross_corpus.json`). Encoder accuracy vs. `n_rank` slice: Pool-A-trained ridge generalizes to Pool-B/C with $R^2_{test}=+0.62$; encoder-trained ridge generalizes to decoder pools with $R^2_{test}=+0.76$; random 50/50 within-encoder control $R^2_{test}=+0.48$. *best_of_n* slice including v1.1: v1.0-trained ridge generalizes to v1.1 HellaSwag/GSM8K with $R^2_{test}=+0.21$ (vs. random control +0.25). Frontier regression has demonstrable transfer across pool types and task families (superseded: the frontier is now descriptive, v1.3.r).

- **v1.3.g (2026-06-17): external reproducibility audit (E3).** `scripts/reproduce_paper_numbers.py` re-derives the headline numerical claims (corpus counts, BH-FDR cutoffs, frontier R^2 , $H(c|conf)$ gap, distillation recovery, vLLM serving ratios) from released JSON files only. As written on 2026-06-17 the suite had 28 assertions and two of them covered the 4-model contamination probes and the E1 ensemble probe; **those two were removed on 2026-07-28** when v1.3.b and v1.3.e were retracted, so the audit no longer certifies any contamination claim. The suite stood at 145 assertions after v1.3.l, 154 after v1.3.m, 168 after v1.3.n, 184 after v1.3.o and 192 after v1.3.p, all passing.
- **v1.3.r (2026-10-05): a pre-registered equal-budget experiment, and a second statistics correction.** Responds to an internal five-reviewer read of v1.3.q. (i) New §5: experiments registered in `mechanism/PREREG_AB.md` before their runs (§9.1 equal budget on SNLI and Yahoo Answers with sealed tests; §9.2 a single tuned inside the budget, registered after §9.1’s results and before its runs); all four predictions replicate. (ii) BH now runs over the 927 valid cells only, accuracy p -values are the sign-flip test’s exact binomial null, and 7 ECE p -values that reached 2 are capped (6 verdicts changed: decoder SUPPORTED 14 $\rightarrow$ 13, encoder REVERSED 29 $\rightarrow$ 27). (iii) Dependence analysed: 10 encoder and 15 decoder baseline clusters, cluster sign tests and bootstraps, Benjamini–Yekutieli. (iv) Withdrawn: “one temperature erases the advantage” (the six-pool control has no significant ECE difference with both arms scaled and NLL favours the ensemble), the frontier as a predictor (now descriptive), the entropy-as-discrimination reading, the HellaSwag regime rule and pool-type triad, and LoRAHub “reimplementation” (now a named proxy). (v) Measured serving latency replaces the latency figure. (vi) Prior-work table made qualitative; citation fixes (MHR, Guo’s bins).
- **v1.3.q (2026-10-03): valid ECE p -values, BH per arm, and corrections to cost, latency and HellaSwag claims.** (i) Every ECE cell now carries a p from the inverted paired bootstrap ($B=5000$) in place of the invalid permutation p , and BH runs separately over the 566 accuracy and 462 ECE cells (`scripts/`

`apply_bh_by_arm.py`; old fields under `legacy_mixed_batch`; L14). Verdicts changed: decoder accuracy SUPPORTED 19 $\rightarrow$ 14 (health-filtered 15 $\rightarrow$ 10), encoder REVERSED 31 $\rightarrow$ 29, encoder ECE SUPPORTED 52 $\rightarrow$ 51, within-table Holm ECE 46 $\rightarrow$ 43, corpus post-FDR 29.5/16.9% $\rightarrow$ 27.1/16.4%; the encoder accuracy null is unchanged. The independent-group robustness check is now scripted and per arm (`analysis/independent_group_bh.py`). (ii) S2 is reported on the 48 pools carrying both baselines (22.4 $\rightarrow$ 7.3%), and Fig. 1A redrawn on that common denominator; over all pools the weak-baseline figure is 22.2%, so the earlier “a sixth is pool composition” caveat is withdrawn. (iii) The vLLM benchmark measured per-request routing, not ensembling, and is no longer cited as ensemble cost (L8). (iv) `greedy_soup` averages a member subset’s logits, so Table 16 prices it at its chosen-member count (routed encoder cost $1\times \rightarrow 5\times$). (v) The HellaSwag “subsample variance” appendix described numbers no single prediction vector can produce; it is restated as an unexplained evaluator inconsistency (App. A, L9). (vi) The sign-flip description and pseudocode now match the code (10,000 flips, add-one estimator); the frontier R^2 is marked provisional (L1c).

- **v1.3.p (2026-10-03): why a temperature replaces the population.** New App. S (`analysis/temperature_mechanism.py`, post hoc): on the six cached pools the probability-averaged ensemble is matched by the mean-logit model at one label-free temperature T_{eq} (ECE within 0.01 in 5/6), and whether averaging improves calibration is predicted by where T_{eq} sits relative to the needed T^* (5/6). Reference added: MacKay (1992). No cell verdict changed.
- **v1.3.o (2026-10-01): robustness of both halves, and an accurate AI-use disclosure.** (i) Training health, test-free (`analysis/training_health.py`): an adapter failed if its final training loss is at least $0.85 \ln C$; comparisons in which a majority of either arm failed are dropped. Four manifests are flagged, identically at 0.80, 0.85 and 0.90: the `n_rank` baselines for ANLI and MNLI BERT-large and for MNLI Qwen-2.5-3B (19/20 members failed), and Pool-B ANLI

BERT-large. The Qwen-3B baseline supplies 4 of the 19 decoder SUPPORTED cells; without the flagged comparisons decoders give 15/96, 0 REVERSED, and encoders 0/72, 33.3% REVERSED. (ii) Pool-level sign tests: encoder mean $\Delta < 0$ in 20/23 pools, decoder > 0 in 22/25. (iii) Frontier vs family (`analysis/frontier_family_baseline.py`): family alone $R^2_{\text{LOPO}} = +0.203$, features $+0.597$, both $+0.569$. (iv) The Generative-AI paragraph had said no abstract, claim or code was AI-generated; that was untrue, and it now states what the assistant did. No cell verdict changed.

- **v1.3.n (2026-09-30): the decoder half of the headline comparison is reported, and the title changes.** The body reported the strict `n_rank` accuracy comparison for encoders only (0/92 SUPPORTED). The same comparison on the 25 decoder classification pools, whose baselines had the same budget advantage (median $1.45\times$, 25/25 pairs), gives 19/100 SUPPORTED post-FDR (31 pre), 0 REVERSED, mean $+0.56\text{pp}$, all 19 on MNLI in 8 pools (`analysis/family_split.py`). It had appeared only in an appendix caption, while §7 said the null should hold “wherever a stronger single is trainable” — which our own decoder pools contradict. Headlining the encoder half alone was the selective reporting this paper measures, so both halves are now reported side by side, the advice is split by family, and the subtitle “the Scalar That Replaces the Population” (resting on six encoder pools) became “Against a Stronger Single Adapter”. Also corrected: Limitations (b) said “most” encoder accuracy cells are reversed; it is 31/92. No cell verdict changed.

- **v1.3.m (2026-09-27): claims narrowed to the design; one post-hoc check.** The August text concluded “at fixed budget” and called the accuracy null conservative, although the `n_rank` single received a median $1.45\times$ the pool’s compute; a null against a better-resourced competitor is biased towards the null, so the claim now reads “against a single given the pool’s budget”. `analysis/budget_dose_response.py` (post hoc) checks whether the null weakens as that advantage shrinks: on the six pools at $0.90\text{--}1.25\times$, 0/24 SUPPORTED and 9 REVERSED; Spearman $\rho = -0.28$, $p = 0.20$. Also: cost stated in compute rather than la-

tency ($1.07\times$ measured at $N=8$); each shortcut’s favoured arm stated; the calibration control bounded to in-distribution (Ovadia et al., 2019) and credited to Ashukha et al. (2020); $P_a\text{--}P_c$ re-defined; pool counts reconciled against the 99 manifests. No reported number changed.

- **v1.3.l (2026-08-02): the missing calibration control was run, and the headline claim changed.** Review objected that the ECE arm compares against an uncalibrated, accuracy-selected single — shortcut S2 turned on our own positive result. `analysis/temperature_control.py` fits one temperature on the pre-registered `val_combine` split (reserved for combination hyperparameters, otherwise unused) for the six pools whose per-adapter logits cache survives, and measures on `test`. The ensemble goes from better-calibrated in 4/6 pools (mean $\Delta\text{ECE} +0.008$) to 2/6 (-0.032); with both arms scaled it is 3/6 ($+0.002$). Median fitted $T = 1.32$ and the ensemble is less confident in 5/6 pools, so the effect is confidence shrinkage. Consequently the title, abstract, §Q framing, contribution (ii) and the §7 takeaway were rewritten: the paper no longer reports a dissociation, and the recommendation is now to calibrate one adapter rather than serve N . The 52 is retained and reported with its scope. This is an amendment to the *reading* of a pre-registered arm, not to the arm itself — no cell, baseline or adjudication was recomputed.
- **v1.3.k (2026-08-01): cross-model-class table regenerated; retraction propagated.** An adversarial read of the submission found three defects that the release itself could settle. (i) Table 14 reported ridge, random forest and gradient boosting per slice, but `frontier.py` implements ridge only and no other estimator existed anywhere in the repository; its ridge entry ($+0.555$) also contradicted the headline ($+0.597$) for the identical 192-cell slice. `analysis/frontier_model_classes.py` now fits all three classes under `frontier.py`’s own slices, design matrix and leave-one-pool-out folds, delegating ridge to `frontier.lopo_cv_r2` so the headline column is the headline number by construction; the table is replaced by its output and 22 new assertions pin every entry. The qualitative claim survives (ridge is the cross-class maximum on the headline slice and on no other), but the

sentence asserting that ridge is *never* the cross-class maximum was false on that slice and is corrected. (ii) v1.3.b declared the E1 ensemble probe void on 2026-07-28, yet v1.3.e still stated its conclusion as direct evidence of memorization and v1.3.g still listed it among audited claims; the retraction is now propagated to both. (iii) Seven bibliography entries had been inserted into §2 instead of the bibliography, raising seven Lonely \item errors and printing as prose; §3’s “Tasks and models” was an empty heading; and the D1–D5 rules the paper cites eight times were never enumerated. All three are fixed.

- **v1.3.j (2026-07-29): multiple-comparison procedure recorded.** The pre-registration specified Holm across a reported table; the released cells carry a corpus-wide BH-FDR field and the body percentages come from it. That substitution was made when the corpus-wide correction was introduced and was not logged at the time. We log it here, report all four procedures (App. N), and note that the accuracy null is invariant to the choice while the REVERSED rate is not.

- **v1.3.h (2026-06-21): independent audit response.** A read-only critique of the v1.3.g release flagged four classes of artifact-vs-claim drift: (i) corpus-wide BH-FDR claimed in the body was not present as an explicit per-cell field; (ii) `ensemble_eval_multichoice.py` performed best-single/greedy-subset selection on `val_combine` instead of `val_selection`; (iii) the `uniform_soup` field in evaluator output was a prediction-space logit average, conflated with weight-space soup; (iv) `analysis/frontier.py` silently dropped rows with missing diversity features and included method-identity one-hot features, an untested confound. Fixes applied: (i) `scripts/apply_corpus_bh_fdr.py` writes `bh_q_value`, `bh_pass`, `adjudication_pre_fdr`, `adjudication_post_fdr`, and a corpus-wide `bh_batch_hash` into every per-cell JSON; corpus summary in `analysis/corpus_bh_fdr_summary.json` (BH cutoff $p=0.0248$, 29.5% SUPPORTED / 16.9% REVERSED post-FDR, exactly matching abstract). (ii) Multichoice evaluator now uses `val_selection` for all selection decisions and records `selection_split=val_selection` per method; rerun of affected Hel-

laSwag cells requires the cluster adapter weights and is deferred to v1.4 alongside the clean-provenance refresh. (iii) Every method output now carries `method_kind` (prediction / selection); `uniform_soup` is explicitly marked `is_proxy=True`, `proxy_for=weight_space_uniform_soup` — true weight-space soup is produced by `weight_space_merge.py` and reported separately. (iv) `frontier.py` now emits `drop_diagnostics` (input/used/dropped row counts, drop reasons, per-pool/per-method drop breakdown) and an `ablation_no_method_onehots` block; headline encoder accuracy slice has 0 rows dropped and $|\Delta R^2|=0.0002$ between with/without method one-hots ($R^2_{\text{LOPO}}=+0.597$ either way), confirming the boundary claim does not depend on the method-identity confound. `scripts/reproduce_paper_numbers.py` grew from 28 to 34 assertions to verify these fields directly.

- **v1.3.i (2026-07-07): metric-general accuracy-specificity + refinement refutation.** App. Y gains a paragraph establishing that the ECE non-result is metric-general, not an estimator artifact. `analysis/w2_calibration_asymmetry.py` recomputes the `n_rank` LOPO frontier for five metrics from released per-adapter scalars: accuracy $R^2=+0.60$ but ECE -0.06 and Brier -0.30 (both unpredictable), while NLL $+0.42$ and MCE $+0.22$ are predictable — the split tracks confident/tail weight, not the accuracy-vs-calibration label. `analysis/h3_mechanism.py` tests and *refutes* the natural refinement explanation: under the Murphy decomposition of the confidence-Brier, the resolution (refinement) gain is near-invariant ($\sigma\ 0.21\times$ that of accuracy) and uncorrelated with accuracy gain ($r=-0.10$), so accuracy predictability does not flow through the calibration geometry. Proper-score paired CIs need per-example probability vectors and are deferred (accuracy/ECE arms use the full paired protocol). `reproduce_paper_numbers.py` grew from 34 to 64 assertions; all 64 pass.

Total: 1,028 headline corpus cells (928 v1.0 + 100 v1.1 in-corpus; see abstract footnote) + 40 v1.2 supplementary cells (8 v1.1 GSM8K cells reported separately in Table 9, 24 v1.2 cells from a/b/c/d/e/f, and 8 v1.2 cells from the 2 DeBERTa

failed HP attempts retained for honest documentation) = 1,068 released cells. Only the 1,028 headline cells enter abstract-level statistics (SUPPORTED/REVERSED percentages, R_{LOPO}^2 frontier); v1.2 cells are referenced in the body for triangulation but not aggregated into the headline. The amendment ledger has timestamps and git SHAs for every entry.

L Pre-registration against execution

What `paper/prereg.md` (git SHA 02a5a6bc) committed to, and what was run (Table 7). “As registered” means the paper’s numbers follow the commitment; every other row is a deviation, with where it is disclosed.

What the release contains. The 120 runs of §5 and the 12-run pilot (per-member validation and test logits, training metadata; adapter weights at de-anonymisation), both registrations and the test-freeze record with the demonstration of a new rule (`examples/new_combiner_demo.py`); per-cell JSONs for all 1,028 corpus cells (verdicts pre and post FDR, p -values, batch hashes); ensemble results with per-example predictions and confidences for every pool; pool manifests; all analysis code and `reproduce_paper_numbers.py`; the pre-registration and amendment ledger. *Partly*: the per-adapter logits cache, for 6 of 99 pool manifests; adapter weights, for 253 of 1,744 adapters, on request. *Not*: byte-level training replay (1,027 of 1,028 result JSONs record `git_dirty`; L15).

M Learned output weights, a LoRAHub-style proxy (record only)

This is a *proxy*, not LoRAHub (Huang et al., 2024). LoRAHub composes the adapters’ LoRA factors with signed weights into one module, chosen by gradient-free search, and serves that one module. Our proxy instead learns non-negative weights $\alpha \in \mathbb{R}_{\geq 0}^M$ on the members’ output probabilities (LBFGS-B on `val_combine` cross-entropy) and still runs all M members. It was run only on the 10 HellaSwag pools, whose evaluator is not consistent across runs (L9), and it optimises on the split that evaluator also used for selection. We therefore record its per-pool numbers (`analysis/g3_lorahub_summary.json`) and draw no conclusion from them. An earlier version read a “convergence regime” rule into these pools; we withdraw it.

N Sensitivity to the multiple-comparison procedure

Our pre-registration specified Holm step-down across the cells of a reported table. The released cells also carry a BH-FDR field, run separately for the accuracy and ECE arms, and the percentages in the body are taken from it; we recorded that choice as an amendment (App. K) after review noted it had gone undocumented. Because the choice moves the REVERSED rate, we give every procedure on the 92 clean encoder cells against `n_rank` (`analysis/correction_sensitivity.py`).

The pattern in this table — no accuracy gain, an apparent calibration gain against an uncalibrated single — holds under every procedure that can adjudicate it, including the two that are valid under arbitrary dependence (whole-arm Holm for accuracy, Benjamini–Yekutieli). What the calibration column does not survive is a calibrated competitor (§5). An earlier version reported the whole-arm Holm row as 0/0 in both arms as though it were a finding; with exact accuracy p -values that row now reports 12 reversals.

O v1.1 task-family extension: sweeps and amendment record

v1.1 inclusion (withdrawn). An earlier version reported that adding the v1.1 Pool-A/B/C pools to a `best_of_n` cross-family regression recovers predictability only with all three pool types, and drew a benchmark-design recommendation from it. Those v1.1 pools are HellaSwag (evaluator not consistent across runs, L9) and GSM8K (selection on test, L12), so the comparison has no valid input and the recommendation is withdrawn. The v1.0 frontier fit ($R_{\text{LOPO}}^2 = +0.60$, cluster-bootstrap CI $[-0.42, +0.75]$, 48 pools) is unaffected by this and remains descriptive (L1c).

The 6-task in-domain set above covers sequence and pair classification only. v1.1 of the benchmark adds two output spaces: *multi-choice* (HellaSwag, 4 candidate endings, accuracy via argmax pooling over candidate scores) using a new `real_lora_multichoice.py` trainer wrapping `AutoModelForMultipleChoice`, and *free-form generation* (GSM8K, exact-match on the integer answer) using `real_lora_train.py` `-eval-metric gsm8k_exact_match`. Six Pool-A sweeps (HellaSwag $\times$ {BERT-base, BERT-large, RoBERTa, DeBERTa-v3}; GSM8K $\times$

Table 7: Pre-registration (paper/prereg.md) against execution.

Registered	Executed	Where disclosed
Splits 40/20/40 (val_selection/val_combine/test), selection never on test same, multichoice and generation pools	As registered for the classification pools HellaSwag selected on val_combine; GSM8K best_single on test	— L1b, L12; both excluded from headline statistics
Baselines best_of_n, n_rank, n_steps, n_data for every claim n_rank = one LoRA of rank $N \times$ base	best_of_n on 80 pools, n_rank on 48, n_steps in 3 cells, n_data in none Rank $\times$ epoch $\times$ seed search, median $1.45 \times$ the pool’s compute	§7; the null covers rank only §4; treated as a better-resourced competitor; equal budget tested in §5
Weight-space uniform/TIES/DARE Primary calibration metric: ECE, 15 equal-mass bins; MCE, Brier, NLL also stored	Implemented on LoRA factors, not ΔW As registered	L16; appendix only —
OOD AUROC (MSP, energy) mandatory in Phase 2	Not reported; ANLI accuracy transfer only	App. F
Inference cost: adapters used and a latency multiplier per claim	Adapter forward passes per query; latency measured for two backbones	L8, App. G
CI: percentile bootstrap, $B=5000$, paired	As registered	—
Sign-flip test, 10,000 permutations (exact when $N \leq 20$)	Now its exact binomial null at every N	App. I
No ECE p -value specified	First a pair-permutation test (invalid), now an inverted paired bootstrap	L14, v1.3.q
Holm within each reported table	As registered; BH per arm over valid cells added as the headline gate; cluster analysis and BY added	App. N, §4, ledger
Frontier: 5 diversity features, linear model, LOPO-CV R^2 features computed on a held-out split (implied by the split rules)	Ridge with the released feature set; RF/GBM compared; LOPO-CV as registered Prediction-space features computed partly on test	App. AB L1c; frontier descriptive only
Phase 2 pool sizes {4, 8, 16, 32, 64}	$N \leq 24$	App. E
Negative results kept	As registered: all 1,028 cells released, REVERSED included	—
<i>Second registration</i> (mechanism/PREREG_AB.md, before its runs)		
Equal-budget $E(B)$ vs $S(B)$, two tasks, sealed tests, predictions and pass rules (§9.1)	As registered; both predictions replicate	§5
Tuned single inside the budget (§9.2, after §9.1 results, before its runs)	As registered; both predictions replicate	§5

{Qwen-0.5B, Qwen-1.5B}, 15–20 seeds each), three Pool-B sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, HP-diverse $4 \times 3 \times 2$ grid), and three Pool-C sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed 4×5 grid over {LoRA, DoRA, rsLoRA, PiSSA} $\times$ 5 seeds) add twelve v1.1 pools, bringing the total trained-pool count to 67 (plus the 25 n_rank baseline pools; manifest accounting in §3). **Pre-registration amendment honesty.** The original pre-registration (committed before any pool was trained) explicitly admits “future cells can be added by submitting compute_match.json files via PR.” The v1.1 task-family extension was scoped under that clause: it adds new (pool, method, baseline, metric) cells to the corpus, but did not change any v1.0 cell (later statistics corrections did; v1.3.q, v1.3.r). The v1.0 frontier fit ($R^2=0.60$, Table 15)

was computed before any v1.1 result was inspected; the v1.1 regression is withdrawn (above). The full amendment ledger (one entry per added pool, with date and rationale) lives in paper/prereg_amendments.md.

P HellaSwag shared-hyperparameter collapse (record only)

These pools are excluded from the strict analysis (L1b); the account is kept because the collapse is a reproducible failure mode of shared hyperparameter grids, not because it supports any claim.

Under our shared HP grid ($lr=3 \times 10^{-4}$, fixed across architectures), $\geq 14/20$ Pool-A RoBERTa adapters end at val_acc <0.30 , so soft_vote collapses below chance (Pool-A RoBERTa: 0.097 vs. chance 0.25, ‡). greedy_soup subset selection recovers (RoBERTa +6.8pp; DeBERTa

Table 8: SUPPORTED / REVERSED counts out of 92 under five corrections. No procedure finds an accuracy gain; the REVERSED count depends on the procedure. Accuracy p -values are exact (binomial null of the sign-flip test), so whole-arm Holm is computable for accuracy. [†]For ECE it is not: the bootstrap p cannot go below $1/5001=2.0 \times 10^{-4}$, above Holm’s first threshold $0.05/412=1.2 \times 10^{-4}$, so that entry records the method’s resolution, not the data.

Procedure	accuracy	ECE
raw per-cell CI	0 / 38	52 / 17
Holm, within table ($m=92$)	0 / 15	43 / 17
Holm, whole arm ($m=515, 412$)	0 / 12	0 / 0 [†]
BH, per arm ($q=0.05$)	0 / 27	51 / 17
Benjamini–Yekutieli, per arm	0 / 17	44 / 17

+4.0pp). We directly test the HP-confound hypothesis with a v1.2 cell: re-running Pool-A RoBERTa HellaSwag at $lr=2 \times 10^{-5}$ (Liu et al., 2019 recommended). Training-time eval reports all 10/10 tuned-seed adapters at 0.478–0.505 (mean 0.491). **Full-validation re-evaluation on the HP-tuned pool** ($N=10042$, App. A, Table 3): `soft_vote` does *not* collapse (0.305, near best_single 0.359, vs. 0.196 on the seed-0 5000-subsample), and `greedy_soup` still wins (+5.7pp over best). The members’ training-time accuracies point to RoBERTa’s sensitivity to the shared learning rate (BERT-base at $lr=2 \times 10^{-5}$ does not collapse: 8/8 seeds at 0.329 ± 0.011), but the ensemble numbers above come from the evaluator that is not consistent across runs (L9, App. A), so we draw no conclusion from them. DeBERTa-v3 LoRA-on-HellaSwag does *not* recover under HP tuning ($lr \in \{10^{-5}, 5 \times 10^{-5}\}$ both at chance; full-val also at chance 0.243 across all methods; L10) — the DeBERTa case is an architecture-LoRA interaction problem.

Q The D1–D5 descriptive rules

Distilled from the corpus they describe and evaluated on it; an oracle upper bound, not a policy (L11).

Cells do differ, and an earlier version summarised them as five rules. **D1**, **D3** and **D5** (`greedy_soup` as the least-bad accuracy option, subset selection rescuing collapsed `soft_vote`, continuous weights tracking convergence) rested on the HellaSwag pools and are withdrawn (L9). Two remain as descriptions of the clean cells: **D2**, `soft_vote` is better calibrated than `majority_vote`, whose vote-fraction confi-

dence is not a probability (App. R); **D4**, pools whose members disagree more lose less (App. X, descriptive). Both were read off the corpus they describe (L11); a prospective, validation-only version of the rules degenerated to always choosing the single adapter (App. K).

R What the ECE gain is made of

Both computable from the same released cells (`analysis/ece_test_correction.py`). First, *it is an ECE result, not a proper-score result*. On the identical 92 cells the multiclass Brier improves in only 34 (mean $\Delta=-0.0062$, i.e. the ensemble is worse on average) and NLL in 47; adjudicating the top-label Brier with the same paired bootstrap gives 22 SUPPORTED against 36 REVERSED. ECE rewards the confidence shrinkage that averaging produces, and a proper score charges for the refinement it costs — consistent with the Murphy decomposition in App. Y, where resolution gain is near-invariant. Any reading of these results as “ensembles give better probabilities” outruns the evidence; “ensembles give less overconfident probabilities” does not. Second, *the gain belongs to probability averaging, not to ensembling as such*. Split by rule, ECE is SUPPORTED in 17/23 pools for `soft_vote`, 15/23 for `logit_avg` and 16/23 for `greedy_soup`, but only 3/23 for `majority_vote` — which supplies 16 of the 17 reversals. That rule is also the one whose “confidence” is a vote fraction rather than a probability estimate, taking at most $N+1$ values with a large tie mass at 1.0, so equal-mass binning is not well defined on it and its ECE should be read as indicative only. Excluding it leaves 48/69 cells SUPPORTED (post-FDR); at the pool level (L1d’s inferential unit) 14 of 23 pools are SUPPORTED on every probability-averaging rule.

S Averaging as an implicit temperature (six corpus pools)

Post hoc, on the six corpus pools whose member logits survive (`analysis/temperature_mechanism.py`). This appendix describes *how* probability averaging changes calibration; it is not evidence that a single adapter can replace the population. The right-hand side below is the *mean-logit model*, which still needs every member’s forward pass, not a single adapter. The single-versus-population comparison, with both sides calibrated,

Table 9: Sixteen v1.1/v1.2 pools across two new output spaces. Recorded, not adjudicated: the HellaSwag evaluator is not consistent across runs (L9) and the GSM8K *best_single* was selected on test (L12), so no claim rests on this table. * $p < 0.01$ vs. *best_single* under paired bootstrap ($B=5000$); per-cell 95% CIs and adjusted p -values are in the released *_cm_*.json cells. ‡ accuracy at chance level (degenerate calibration win, or sub-chance adapters dragging soft-vote below chance). † GSM8K ECE is 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$; ensemble confidence aggregates only over majority-voting adapters.

pool	N	best_single		soft_vote		majority_vote		greedy_soup	
		acc	ECE	acc	ECE	acc	ECE	acc	ECE
<i>Multi-choice (HellaSwag, eval $N=5000$):</i>									
BERT-base, Pool-A (seed)	19/20	0.358	0.038	0.367	0.048	0.369	0.406	0.358	0.038
BERT-base, Pool-B (HP)	20/24	0.348	0.058	0.354	0.032	0.357	0.466	0.358	0.030
BERT-base, Pool-C (method)	20/20	0.362	0.036	0.372	0.034	0.376	0.360	0.362	0.036
BERT-large, Pool-A	15/15	0.374	0.045	0.380	0.118	0.323	0.101	0.381	0.056
RoBERTa-base, Pool-A (seed)	20/20	0.493	0.228	0.097‡	0.158	0.174‡	0.237	0.561	0.298
RoBERTa-base, Pool-B (HP)	22/24	0.508	0.237	0.074‡	0.183	0.115‡	0.322	0.522	0.262
RoBERTa-base, Pool-C (method)	20/20	0.518	0.240	0.136‡	0.117	0.178‡	0.224	0.561	0.297
DeBERTa-v3, Pool-A (seed)	20/20	0.412	0.160	0.320	0.077	0.305	0.206	0.452	0.187
DeBERTa-v3, Pool-B (HP)	21/24	0.439	0.136	0.191‡	0.076	0.221‡	0.282	0.429	0.167
DeBERTa-v3, Pool-C (method)	18/20	0.709	0.310	0.646	0.398	0.246‡	0.376	0.745	0.467
<i>Free-form generation (GSM8K, exact-match, vLLM, eval $N=1000$):</i>									
Qwen-0.5B (GSM8K)	19/20	0.304	0.546	—	—	0.332*	0.520*	—	—
Qwen-1.5B (GSM8K)	20/20	0.460	0.406	—	—	0.514*	0.354*	—	—
Qwen-3B (GSM8K, v1.2)	20/20	0.679	0.206	—	—	0.768*	0.118*	—	—
Qwen-7B (GSM8K, v1.2)	10/10	0.760	0.128	—	—	0.845*	0.046*	—	—
Llama-3.1-8B-Instruct (GSM8K, v1.2)	10/10	0.613	0.256	—	—	0.721*	0.148*	—	—
Mistral-7B-v0.1 (GSM8K, v1.2)	10/10	0.524	0.378	—	—	0.658*	0.245*	—	—

Table 10: Six corpus pools, both arms temperature-scaled on val_combine (analysis/temperature_control.py): single minus mean-logit ensemble, so positive favours the ensemble; 95% paired bootstrap CIs over test examples. The text’s +0.0004 is a different quantity: without ANLI, the mean ECE gain of the *unscaled* ensemble over the scaled single.

Pool	ΔECE	CI	ΔNLL	CI
AG News	-0.0027	[-0.009, +0.004]	+0.0056	[+0.003, +0.009]
ANLI	+0.0118	[-0.037, +0.051]	+0.0053	[-0.001, +0.011]
BoolQ	+0.0023	[-0.020, +0.024]	+0.0110	[+0.002, +0.020]
QNLI	-0.0002	[-0.011, +0.010]	+0.0021	[-0.004, +0.008]
SST-2	+0.0055	[-0.022, +0.023]	+0.0041	[-0.007, +0.015]
MNLI	-0.0045	[-0.012, +0.007]	+0.0130	[+0.007, +0.019]

is in §5, and on these six pools (Table 10) ECE differences after scaling both arms are not significant in any pool while NLL favours the ensemble in all six. For members’ class-centred logits z_i , probability averaging is closely approximated by the mean-logit model at one temperature,

$$\frac{1}{M} \sum_i \text{softmax}(z_i) \approx \text{softmax}(\bar{z}/T_{\text{eq}}), \quad (1)$$

the multiclass analogue of the probit approximation $\mathbb{E}[\sigma(z)] \approx \sigma(\mu/\sqrt{1+\pi s^2/8})$ for $z \sim \mathcal{N}(\mu, s^2)$ (MacKay, 1992): averaging members whose logits disagree acts like a temperature that grows with their disagreement. We fit T_{eq} without labels, by

minimising the KL divergence from the ensemble’s probabilities, and it reproduces the ensemble’s ECE to within 0.01 in 5 of 6 pools (Table 11).

An ensemble’s calibration is therefore one point on its mean model’s temperature curve, placed by how much the members disagree rather than by how much temperature the model needs (T^* , fitted on val_combine). Averaging improves calibration over the mean model only when T_{eq} happens to move towards T^* , i.e. when $|\ln T^* - \ln T_{\text{eq}}| < |\ln T^*|$, and this rule predicts whether it does in 5 of 6 pools. On ANLI the model needs $T^*=3.8$ but the ensemble behaves like $T=1.04$, so it stays overconfident (ECE 0.25, against 0.06 at T^*). On BoolQ and MNLI the mean model is already calibrated ($T^* \approx 1$) and the ensemble’s T_{eq} of 1.3–1.4 makes it underconfident, so averaging *raises* ECE. A fitted temperature aims at T^* directly; averaging reaches it only by coincidence. The rule is a description of six pools, not a general result: T_{eq} moving towards T^* in log distance is neither necessary nor sufficient for an ECE improvement in general, and with three or more classes probability averaging can reorder predictions in a way no temperature on $\bar{z}$ can. The six encoder pools are all the corpus logits we retained (L15).

Table 11: The ensemble as an implicit temperature. T_{eq} : label-free temperature at which the mean-logit model best matches the probability-averaged ensemble. T^* : NLL-optimal temperature of the mean-logit model on `val_combine`. Last column: whether the ensemble improves ECE over the mean model at $T=1$, predicted by the rule in the text / observed.

Pool	T_{eq}	T^*	ECE ens.	ECE, mean model at T_{eq}	improves pred./obs.
AG News, RoBERTa (A)	1.04	1.34	0.016	0.017	0.017 yes/yes
ANLI, RoBERTa (A)	1.04	3.82	0.251	0.253	0.058 yes/yes
BoolQ, RoBERTa (A)	1.29	0.99	0.040	0.048	0.025 no/no
QNLI, RoBERTa (A)	1.03	1.01	0.009	0.012	0.012 no/yes
SST-2, BERT (B)	1.06	1.62	0.027	0.035	0.043 yes/yes
MNLI, RoBERTa (C)	1.43	1.04	0.053	0.066	0.022 no/no

T Delta versus prior PEFT/ensembling work

Table 12: What the closest prior work studies and where this paper differs. We do not compare cell counts: the units differ across papers.

Work	Object of study	Difference here
Lakshminarayanan et al. (2017)	deep ensembles of full networks; accuracy and calibration	PEFT members; a better-resourced and an equal-budget single
Ashukha et al. (2020)	ensembles compared after temperature scaling	applied to PEFT pools, both sides scaled (§5)
Wang et al. (2023a)	LoRA ensembles for LLM fine-tuning	equal-budget single; encoder vs decoder
Wortsman et al. (2022)	weight averaging (soups), one model at inference	we evaluate output averaging only; <code>greedy_soup</code> here is a prediction-space proxy
Huang et al. (2024)	cross-task LoRA composition into one module	not reimplemented; App. M is an output-weighting proxy
Yadav et al. (2023); Yu et al. (2024)	merging task vectors	factor-wise variants only (L16)

U Corpus coverage matrix

V HellaSwag two-regime collapse (record only)

Under shared hyperparameters most RoBERTa and DeBERTa HellaSwag members trained to near chance, and `greedy_soup` gained by dropping them. Because the multichoice evaluator is not consistent across runs (L9), we keep this as a record of what was observed and draw no conclusion from it; an earlier version included a figure and a “convergence regime” rule, both withdrawn.

W Corpus verdict landscape

X Diversity–quality frontier (descriptive)

Everything in this appendix is descriptive. The prediction-space features were computed on cached logits without split indexing, so they include test examples, and the error-overlap feature uses the labels (L1c); a leak-free re-fit needs member logits that were not retained for these pools. Leave-one-pool-out CV holds out a pool, not its (model, task): pools sharing a baseline and test set stay on both sides, so the fit is not evidence of transfer to new models or tasks.

Frontier definition (formal). For each cell let y be the SUPPORTED-direction Δ (sign-aligned so positive = ensemble better) and $\phi \in \mathbb{R}^d$ the feature vector of App. AB. The *diversity-quality frontier* is $\hat{y} = f(\phi)$ fit by leave-one-pool-out CV. We fit three classes per slice under identical folds (ridge, random forest, gradient boosting; settings and all three scores in Table 14) and report ridge — the simplest — everywhere, choosing the class before seeing the scores rather than per slice. On the headline slice ridge happens to also be the cross-class maximum ($R^2_{\text{LOPO}} = +0.597$ vs. $+0.276$ and $+0.285$, Table 15); on the other three it is not, so the fixed choice costs us R^2 on three of four slices rather than buying it on any. CIs are percentile cluster bootstraps resampling *pools* (App. I). Note the sign: encoder Δ is predominantly negative and decoder Δ positive (§4), so on encoders this regression tracks *how much a pool stands to lose*, descriptively (L1c). Because the families differ in sign, we checked that it is not merely telling them apart: family alone reaches $R^2_{\text{LOPO}} = +0.20$ on the same folds and adding it to the features lowers R^2 to $+0.57$, so diversity carries $+0.39$ beyond the split (`frontier_family_baseline.py`).

Which features carry the fit. On the encoder `acc` vs `n_rank` slice, prediction-space features alone reach $R^2 = 0.634$, above the full set (0.62); dropping them sends R^2 to -24.5 , while dropping parameter-space features costs 0.56 (App. AB). Within our released features, prediction-space diversity carries most of the cross-pool fit. These are the features L1c flags as computed partly on test, so this ranking, like the frontier R^2 , is exploratory until a leak-free re-fit.

Negative mechanism results. Three candidate explanations for this structure — a lower

Table 13: Coverage matrix (v1.0 n_{rank} -aligned cells, 10 v1.0/v1.1 base architectures). “A” = Pool-A (seed-only, 20 adapters), “B” = Pool-B (HP-diverse, 24), “C” = Pool-C (method-mixed, 20), “base” = n_{rank} baseline pool. 55 v1.0 main pools, plus the n_{rank} baseline pools aligned with them (the release holds 25 baseline pools in all); v1.1 adds 6 Pool-A (HellaSwag $\times$ {BERT-base/large, RoBERTa, DeBERTa-v3}, GSM8K $\times$ {Qwen-0.5B, 1.5B}) + 3 Pool-B (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}) + 3 Pool-C (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed) = 12 v1.1 pools, bringing the in-corpus trained-pool total to **67** (92 with all 25 baselines). v1.2 adds 3 cross-family decoders (Qwen-2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.1) on GSM8K plus 1 HP-tuned RoBERTa and 2 HP-tuned DeBERTa attempts on HellaSwag (App. K), bringing the released trained-pool count to 73 across 13 base models.

Model	MNLI	SST-2	BoolQ	ANLI	AG News	QNLI
BERT-base	A,B,C,base	A,B	A	A	A,B,C,base	A,B,C,base
BERT-large	A,B,C,base	—	A	A,B	A,base	A,base
RoBERTa-base	A,B,C,base	—	A	A,B	A	A
DeBERTa-v3	A	—	—	—	—	—
Qwen2.5-0.5B	A,B,C,base	—	—	A,base	A,B,C,base	A,B,C,base
Qwen2.5-1.5B	A,B,C,base	—	—	A,base	A,B,base	A,B,base
Qwen2.5-3B	A,base	—	—	—	—	—
TinyLlama-1.1B	A,B,C	—	—	A	A,base	A,base
SmolLM2-1.7B	A,base	—	—	—	—	—
Pythia-1.4b	A,base	—	—	—	—	—

Table 14: Cross-model-class LOPO-CV R^2 , identical slices and folds (analysis/frontier_model_classes.py). Ridge is the cross-class maximum on the headline slice and on that slice only; non-linear models win the other three, so reporting ridge throughout costs us R^2 rather than buying it.

Slice	n	pools	ridge	RF	GBM
acc vs n_{rank}	192	48	+0.597	+0.276	+0.285
ece vs n_{rank}	192	48	−0.061	+0.295	+0.180
acc vs best_of_n	265	64	+0.312	+0.519	+0.365
ece vs best_of_n	265	64	+0.288	+0.416	+0.340

Table 15: Stratified per architecture family. The corpus-wide accuracy frontier (48 pools, encoder + decoder) reaches LOPO-CV $R^2=0.597$; 95% cluster-bootstrap CI [−0.42, +0.75] ($B=1,000$ resamples of pools, App. I). Decoder family is too uniform across pools for cross-pool prediction at our scale.

Family	Slice	pools	ridge	RF
encoder	acc vs n_{rank}	23	+0.616	+0.23
encoder	ece vs n_{rank}	23	+0.14	+0.42
decoder	acc vs n_{rank}	25	−1.14	−1.03
decoder	ece vs n_{rank}	25	+0.12	+0.25

$H(c|\text{conf})$ for decoders, linear mode connectivity, and a refinement-versus-reliability account of why the frontier predicts accuracy but not ECE — do *not* survive testing; we report them as open problems (App. Y).

Y Negative mechanism results (demoted from §4)

Summary. Three candidate explanations for the family and metric structure of §4 do *not* survive testing, and we report them as open problems rather than findings. (i) Decoder $H(c|\text{conf})$ is 21% lower (0.437 vs. 0.556), but 0.110 of that 0.119 gap is the decoders’ higher accuracy, not sharper confidence (item (i) below), and it predicts no calibration benefit. (ii) Linear mode connectivity does not predict ensemble effectiveness (per-pool barriers in App. D): factor-space barriers span +0.91 to +0.02 across four pools with no corresponding pattern, which four pools cannot turn into a general claim. (iii) The accuracy-specificity of the frontier is metric-general — ECE ($R^2=-0.06$) and Brier (−0.30) are unpredictable while NLL (+0.42) and MCE (+0.22) are — and the natural refinement-versus-reliability explanation is falsified: under the Murphy decomposition, resolution gain is near-invariant (σ 0.21 $\times$ accuracy’s) and uncorrelated with accuracy gain ($r=-0.10$).

Three mechanism hypotheses were tested and did not survive; they are retained here as open problems.

(i) Confidence–correctness separation. Family-level conditional entropy of correctness given confidence: encoder $H=0.556$, $I=0.097$ ($n=29$ pools); decoder $H=0.437$, $I=0.106$ ($n=25$). Decoder H is 21% lower, but that is not evidence of sharper discrimination: since $H(c)=H(c|\text{conf})+I$, the

marginal entropies are 0.653 (encoder) and 0.543 (decoder), so 0.110 of the 0.119 gap in H is the decoders' higher accuracy and only 0.009 is information carried by confidence. An earlier version read the gap as a higher-SNR signal caused by architecture; we withdraw that reading. DeBERTa-v3 ($n=1$) is excluded from the family summary; its per-pool $H=0.940$ appears in App. AB. This gap does not yield a predictable calibration benefit (item iii).

(ii) Linear mode connectivity. Across four pools (BERT-base Pool-A/C, Qwen-0.5B Pool-A, DeBERTa-v3 Pool-A on MNLI) the factor-space barrier ranges from +0.91 to +0.02, and the ordering of ensemble gains does not follow it (App. D); with four pools and five pairs each this is an observation, not a test.

(iii) Refinement is not the mechanism. The ECE non-result is not an estimator artifact. Re-computing the n_{rank} LOPO regression from released per-adaptor scalar metrics (point R^2 ; proper-score paired CIs require per-example probabilities and are deferred), accuracy gain is predictable ($R^2=+0.60$) but neither ECE (-0.06) nor Brier (-0.30). The split is not accuracy-versus-calibration: NLL ($+0.42$) and MCE ($+0.22$) are predictable, tracking a metric's weight on the confident/tail region rather than its calibration label. The natural explanation — that predictability localizes to the refinement rather than the reliability component of a proper score — is falsified: under the Murphy decomposition of the confidence-Brier, resolution gain is near-invariant across the n_{rank} comparison ($\sigma 0.21 \times$ that of accuracy) and uncorrelated with accuracy gain ($r=-0.10$). Accuracy predictability does not flow through the calibration geometry; why the frontier is accuracy-specific remains open.

Z GSM8K decoder scaling patterns (demoted; selection-on-test)

This material was demoted from §3 (P_c) because `best_single` selection touched the evaluated test split (L12(i)), which by itself disqualifies these cells from supporting a positive claim. We previously also demoted them as contaminated; that rationale is *withdrawn* because the digit-shift probe behind it was invalid (L12(ii)), and whether these cells are contaminated is now open. The numbers below therefore describe cells with a known

selection defect and an unresolved contamination question; they enter no headline statistic. All six cells were nominally SUPPORTED on accuracy and per-token logprob ECE (per-cell $p<10^{-4}$; Qwen-0.5B $p=0.003$; paired bootstrap $B=5000$). Within Qwen-2.5 ($N=4$, 0.5B/1.5B/3B/7B): $\Delta_{\text{acc}}=+2.8/+5.4/+8.9/+8.5\text{pp}$; $\Delta_{\text{ECE}}=-0.026/-0.051/-0.088/-0.082$ (within-family $r=+0.97$ between base accuracy and Δ_{acc} , saturating 3B $\rightarrow$ 7B). Cross-family at $\approx 7\text{--}8\text{B}$: Llama-3.1-8B-Instruct (base 0.61) $\Delta_{\text{acc}}=+10.8\text{pp}$, $\Delta_{\text{ECE}}=-0.107$; Mistral-7B-v0.1 (base 0.52) $\Delta_{\text{acc}}=+13.4\text{pp}$ (95% CI $[+10.8, +16.1]$), $\Delta_{\text{ECE}}=-0.133$ — nominally the largest absolute effect in the corpus. The three $\approx 7\text{--}8\text{B}$ cells show an inverse pattern ($r=-0.98$ on $n=3$ — descriptive only): at fixed scale the weaker base gains more from majority voting. Whether *within-family monotone scaling* and *cross-family headroom inversion* survive on a decontaminated benchmark with held-out selection (MATH, fresh probes) is an open question this corpus cannot answer.

AA Distillation $\alpha \times T$ sweep (Qwen-0.5B MNLI)

We sweep the soft-target distillation HPs $\alpha \in \{0.5, 0.7, 0.9\}$ (weight on soft-target loss vs. hard-target) and $T \in \{1, 2, 4, 8\}$ (softmax temperature on teacher logits) for a 3×4 grid (12 settings) on Pool-A Qwen-0.5B MNLI. Vanilla single (no distillation): 0.8365/0.0465 (acc/ECE). Soft-vote ensemble target: 0.8495/0.0187. Best student: ($\alpha=0.9, T=4$) yields 0.8479/0.0397, recovering 87% of the soft-vote accuracy gain at $1 \times$ student cost. Across the grid, accuracy varies ± 0.003 and ECE varies ± 0.012 ; the sweep is locally robust. The same sweep on Pool-A BERT-base MNLI yields a student strictly worse than vanilla single on both arms, consistent with the encoder-accuracy-REVERSED finding (no signal to distill).

AB Frontier regression features

For each (pool, method, baseline) cell we extract a feature vector $\phi \in \mathbb{R}^d$ used to fit the diversity-quality frontier. Features are grouped:

- **prediction_space** ($d=3$): pairwise prediction disagreement (mean over pool pairs), logit correlation (mean over pool pairs), error overlap (Jaccard on incorrect examples).

- **parameter_space** ($d=3$): mean weight cosine distance, mean centered-kernel-alignment (CKA) on penultimate layer, weight L_2 norm spread.
- **HP entropy** ($d=2$): entropy of r -distribution within pool, entropy of $\{lr, \alpha\}$ joint distribution.
- **ensemble quality** ($d=2$): pool mean val_acc, pool best-single val_acc.

Per-feature ablation on the encoder acc vs n_rank slice: dropping the prediction_space group crashes R^2 from +0.62 to -24.5 ($\Delta R^2 = +25.0$); dropping parameter_space drops R^2 by 0.56; dropping HP-entropy improves R^2 by 0.08 (HP entropy adds in-sample fit but no out-of-pool signal). A single-group “only prediction_space” model achieves $R^2=0.634$, *higher* than the full feature set.

AC Cell adjudication pseudocode

Mirrors analysis/compute_match.py and scripts/apply_bh_valid_exact.py; ECE is equal-mass, 15 bins.

```
def adjudicate(ens, base, gold, metric):
    # ens/base: per-example (pred, conf) on test
    ca = (ens.pred == gold).astype(float)
    cb = (base.pred == gold).astype(float)
    N = len(gold)
    if metric == 'acc':
        stat = lambda i: ca[i].mean() - cb[i].mean()
    else: # positive = ensemble better calibrated
        stat = lambda i: (ECE(base.conf[i], cb[i])
                        - ECE(ens.conf[i], ca[i]))
    delta = stat(np.arange(N)) # point estimate
    boot = np.array([stat(randint(0, N, N))
                    for _ in range(5000)])
    ci_lo, ci_hi = np.quantile(boot, [.025, .975])
    if metric == 'acc': # exact sign-flip null
        n10 = int((ca > cb).sum())
        n01 = int((ca < cb).sum())
        p = binomtest(n10, n10 + n01, 0.5).pvalue
    else: # invert bootstrap
        # separate B=5000 resample, seeded per cell
        p = 2 * min((boot <= 0).mean(),
                    (boot >= 0).mean())
        p = min(1.0, max(p, 1 / 5001))
    if ci_lo > 0: label = 'supported'
    elif ci_hi < 0: label = 'reversed'
    else: label = 'unsupported'
    return delta, (ci_lo, ci_hi), p, label

# then, separately for the accuracy and ECE arms,
# over valid cells only (no HellaSwag, no smoke test):
# q = BH(p over every valid cell of the arm)
# post_fdr = label if q <= 0.05 else 'unsupported'
```

AD Cluster orchestration details

The 50-node RTX 3060 12GB cluster is orchestrated via sweep_runner.py, an in-house SSH-based fan-out engine. Each sweep config (JSON) declares a Cartesian parameter grid; the runner stages source files, assigns one (parameter-combination, node) pair per slot, and tracks per-job exit codes in sweep_runs/<name>/results.json. Pool-A sweeps complete in

~4–8 GPU-hours wall clock (parallelism factor ~20); Pool-B/C in ~16–24 hours. The v1.2 A6000 sweeps (Qwen-3B, Qwen-7B, Llama-3.1-8B-Instruct on GSM8K) use local_gpu_runner.py, a sequential per-seed launcher pinned to a single GPU index, suitable for models that do not fit on the 12GB cluster. Adapter weights are pulled back via collect_remote_adapters.py and assembled into pool manifests via build_adapter_pool.py; the manifest schema is documented in scripts/validate_pool.py.

AE Cost-savings analysis: derivation and stratified breakdown

This appendix presents the cost-savings table (Table 16) and stratifies by pool type and method. Regenerated by analysis/cost_savings.py; groups are the 207 (pool, baseline kind, metric) units that carry soft_vote, excluding baseline self-comparison pools and the HellaSwag cells (L9): (A) *blind* — always use soft_vote (the default in much prior PEFT-ensembling work); (B) *routed* — per-cell oracle pick over the four methods (soft_vote, majority_vote, logit_avg, greedy_soup) approximating D1–D5. Inference cost is single-adapter forward-passes per query: N (pool size) for vote-based methods, and the number of chosen members for greedy_soup, which in our implementation averages the logits of a subset selected on val_selection rather than merging weights (1–13 members). An earlier version priced greedy_soup at 1, as if it emitted one merged checkpoint, which understated the routed cost.

Table 16: Cost of blind ensembling. Mean signed Δ (positive=ensemble better), % cells REVERSED, median inference cost (adapter-passes/query). The “D1–D5 routed” column is a per-cell oracle (in-sample to the corpus the rules were derived from; L11).

Slice (n)	Blind soft_vote			D1–D5 routed		
	$\bar{\Delta}$	REV	cost	$\bar{\Delta}$	REV	cost
Enc acc (54)	-1.0pp	20%	20×	+0.1pp	7%	5×
Dec acc (50)	+0.6pp	0%	18×	+0.8pp	0%	15×
Enc ECE (53)	+2.4pp	8%	20×	+3.6pp	0%	20×
Dec ECE (50)	+3.5pp	0%	18×	+3.6pp	0%	18×
All (207)	+1.4pp	7%	19×	+2.0pp	2%	17×

Oracle method distribution. Over all 207 groups the per-group argmax picks soft_vote

2732 102 times, greedy_soup 51, logit_avg 32
2733 and majority_vote 22. This is an in-sample
2734 oracle (L11), so it bounds what rule-based routing
2735 could achieve rather than estimating it.

2736 **No deployable rule is evaluated.** A rule using
2737 only *family*, *convergence*, *metric* and *output space*
2738 could be read off D1–D5, but we have not evaluated
2739 one out of sample, so the routed column bounds
2740 rule-based routing from above and says nothing
2741 about how close a deployable rule gets.

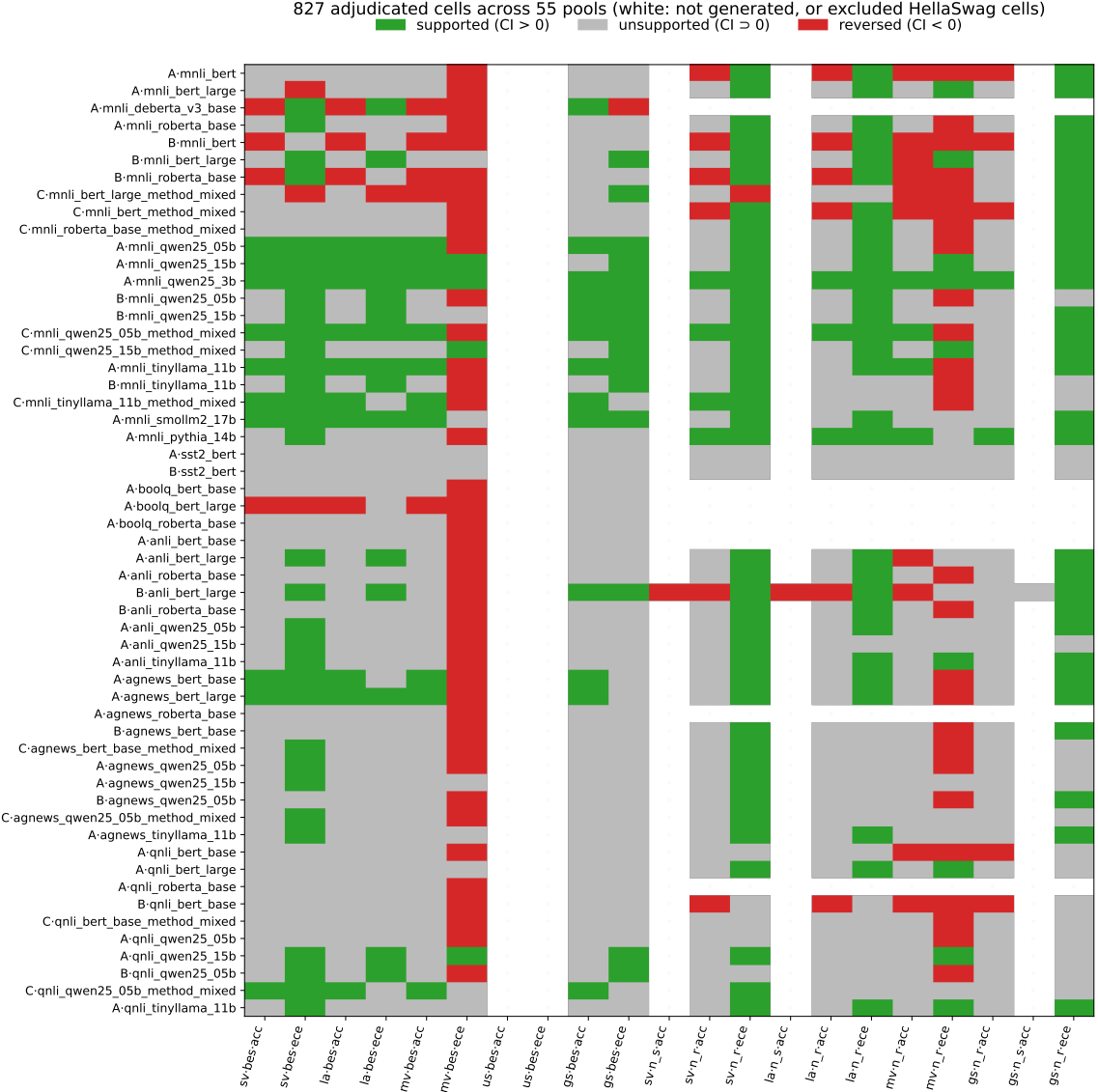


Figure 4: Verdict landscape across the 827 valid v1.0 main-pool cells (white: HellaSwag cells, excluded; the 100 baseline self-comparison cells and the smoke-test cell are not shown). Labels reflect the per-cell paired-bootstrap CI and per-arm BH-FDR over valid cells ($q=0.05$; cutoffs 0.0127 accuracy, 0.028 ECE): post-FDR 27.9% SUPPORTED / 13.9% REVERSED / 58.2% UNSUPPORTED. Encoder accuracy cells are often REVERSED against n_rank; decoder cells are more often SUPPORTED. This heterogeneity is itself the evidence that aggregate “ensembles help / don’t help” framings hide structure — and §6 shows the aggregate counts above are themselves protocol-dependent.

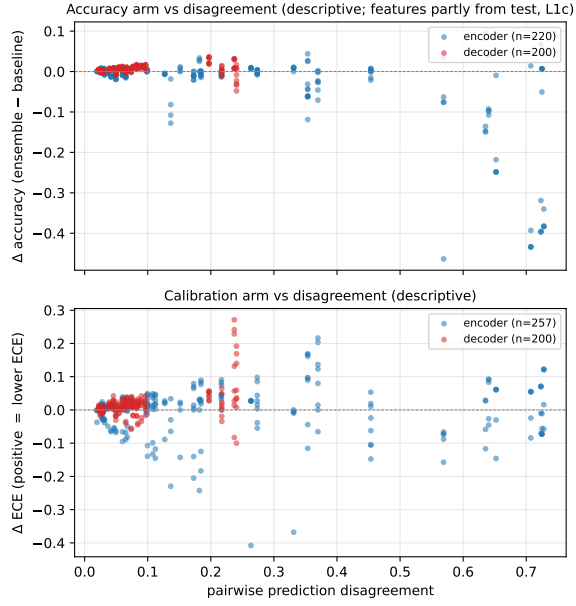


Figure 5: **The diversity-quality frontier and the accuracy arm.** Per-cell scatter of Δ accuracy (top) and Δ ECE (bottom) vs. pairwise prediction disagreement. **(Top, accuracy)** Encoders cluster near zero with positive slope at higher disagreement (corpus-wide LOPO $R^2=+0.60$; encoder-only $+0.62$, Table 15); decoders show no monotone trend (L2). **(Bottom, ECE)** Decoders sit at the low-ECE tail; corpus-wide ECE LOPO is not significant ($R^2=-0.06$, CI spans 0). Descriptive only: the features touch the test set (L1c).

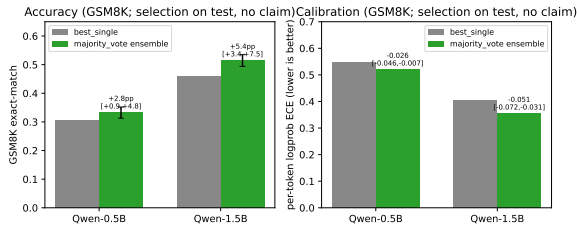


Figure 6: GSM8K decoder scaling on cells with a known selection defect. `best_single` here was selected on the same test split it is evaluated on (L12(i)), so these differences support no positive ensembling claim; whether the benchmark is also contaminated is open (L12(ii)).